

COMMERCIAL STANDARD DIGITAL BUS (CSDB)



GAMA

COMMERCIAL STANDARD

DIGITAL BUS (CSDB)

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prepared by
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PREFACE

This specification, which details the design and implementation of a Commercial Standard Digital Bus (CSDB), is intended to provide the reader with a basic understanding of the data bus and its usage. The specification will be updated as required to reflect changes that occur in its usage or application.

Changes that may occur in the CSDB Design Specifications, will be coordinated through GAMA. New applications and implementations of CSDB should be submitted to GAMA for inclusion into the respective CSDB specifications by the primary system integrator.

This specification pertains only to the implementation of CSDB as used in an integrated system. Overall systems design, integration, and certification remain the responsibility of the systems integrator.

Any changes, comments, or corrections should be forwarded to:

General Aviation Manufacturers Association
Washington, D.C.
Attn: Data Bus Committee

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1. GENERAL

1.1. SCOPE

THIS DOCUMENT IS THE RESULT OF EFFORTS BY GENERAL AVIATION MANUFACTURERS ASSOCIATION TO ESTABLISH & DEFINE THE STANDARDS USED IN THE TRANSMISSION OF SERIAL DIGITAL DATA FOR AVIONICS EQUIPMENTS. BOTH SYSTEM AND ELECTRICAL STANDARDS ARE CONTAINED IN THIS DOCUMENT. EQUIPMENT ADHERING TO THESE STANDARDS IS NOW IN PRODUCTION AT VARIOUS MANUFACTURERS, AND EQUIPMENTS EXPECTED TO INTERFACE WITH THOSE UNITS SHOULD BE DESIGNED TO MEET THIS STANDARD. ONLY THOSE UNITS WHOSE STATUS IS FIRM ARE INCLUDED HERE: THOSE STILL BEING DEFINED ARE DOCUMENTED IN A WORKING DRAFT, COPIES OF WHICH ARE AVAILABLE DIRECTLY FROM THE MANUFACTURER.

1.2. CHANGES

1.2.1. CHANGE MARKINGS

CHANGES TO THIS DOCUMENT FROM THE VERSION IMMEDIATELY PRECEDING THE CURRENT ONE WILL BE MARKED LINE-BY-LINE WITH A BLACK BAR IN THE LEFT MARGIN. SUBSEQUENT VERSIONS WILL DELETE EARLIER CHANGE MARKINGS. PAGES WITH CHANGES OF CONSEQUENCE WILL ALSO HAVE A NEW DATE AT THE BOTTOM.

1.2.1.1 ADDENDUMS (PINK SHEET REVISIONS)

REVISIONS MADE TO THE DOCUMENT IN BETWEEN THE NORMAL REVISIONS TO ENTIRE DOCUMENT (AS DESCRIBED IN SECTION 1.2.1.) WILL BE PRINTED ON PINK PAPER AND THE CHANGES MARKED LINE-BY-LINE WITH BLACK BARS IN THE LEFT MARGIN. OTHER CHANGE MARKINGS ON THE ADDENDUM PAGE WILL BE DELETED TO PREVENT CONFUSION.

1.2.2. HIGHLIGHTS, THIS EDITION

CHANGED: PARAGRAPHS 1.2.1, 1.2.2, 2.1.4, 4.1, 4.2, 4.3.1, 4.3.4, 4.3.5, 4.3.6, AND 4.3.7
ADDED: PARAGRAPH 1.2.1.1

CHANGED: MINOR CORRECTIONS ON ADDRESSES 10, 11, 24, 25, 26, 27, 31, 34, 37, 38, 41, AD, B3, AND C0

ADDED: NEW ADDRESSES (FOR 8.33 KHZ DATA WORDS) 12, 13, AND 35
RENUMBERED PAGES 26 TO END OF DOCUMENT

2. SYSTEM STANDARDS

2.1. DIGITAL BUS DEFINITION

2.1.1. APPLICABILITY

CIRCUITS WHICH ARE DEDICATED TO TRANSMISSIONS BETWEEN ELEMENTS OF A SUBSYSTEM AND WHICH HAVE NO POTENTIAL FOR FUTURE INTERFACES ELSEWHERE IN THE AIRCRAFT MAY DEVIATE FROM THIS STANDARD. SUCH DEVIATIONS SHALL BE DEFINED IN THE EQUIPMENT SPECIFICATIONS, AND SHALL BE SUBJECT TO APPROVAL IN THE FORMAL PRODUCT DESIGN REVIEW.

2.1.2. BUS DESCRIPTION

THE STANDARD METHOD OF DATA TRANSMISSION IS ASYNCHRONOUS BROADCAST DATA IS TRANSMITTED OVER AN INTERCONNECTING CABLE USING DEVICES COMPLYING WITH ELECTRONIC INDUSTRIES ASSOCIATION STANDARD EIA RS-422A.

COMMENT: THE INTENDED CABLE IS A TWISTED, SHIELDED PAIR, USED IN THE UNTERMINATED CONFIGURATION AS DESCRIBED IN RS-422A. TESTING HAS BEEN COMPLETED WITH LINE LENGTHS OF UP TO 50 METERS.

2.1.3. BUS RECEIVER DESIGN PRECAUTIONS

THE FOLLOWING PRECAUTIONS ARE IMPLIED IN THE PARAGRAPHS BELOW, BUT ARE STATED EXPLICITLY HERE AS A RESULT OF SOME EARLY EXPERIENCES WITH IMPLEMENTATIONS.

- A. SEQUENCE SENSITIVITY: THE BUS STRUCTURE OF THIS SPEC IS DESIGN-ED TO USE ADDRESS RECOGNITION EXCLUSIVELY TO IDENTIFY DATA ONCE SYNC IS ACHIEVED, AND TO BE TOTALLY INSENSITIVE TO SEQUENCE. SEE COMMENT UNDER MESSAGE BLOCK DEFINITION PARAGRAPH.
- B. RECEIVER SATURATION LIMIT. EACH RECEIVER MUST BE ABLE TO RECEIVE AT THE MAXIMUM DEFINED BUS RATE WITHOUT SATURATION PROBLEMS, EVEN THOUGH CURRENTLY ASSIGNED ADDRESSES MAY BE FEW ON A GIVEN BUS. OTHERWISE, THE GROWTH CAPABILITY WILL BE LIMITED, AND INTERCHANGEABILITY WILL BE SEVERELY AFFECTED.
- C. RECEIVER INPUT IMPEDANCE IS EXPECTED TO BE NOT LESS THAN 4K OHMS TO ALLOW AS MANY AS 10 RECEIVERS ON A GIVEN BUS.

2.1.4. BUS OPERATIONS

THERE ARE (3) GENERAL FORMS OF BUS TRANSMISSIONS ALL USING THE BYTE, BLOCK, AND SYNC DEFINITIONS OF LATER PARAGRAPHS:

- (1) CONTINUOUS REPETITION: EACH MESSAGE IS UPDATED OR REFRESHED AT THE LISTED INTERVALS. THIS IS THE STANDARD FORM FOR PRIMARY DATA SUCH AS ATTITUDE OR AIR DATA.
- (2) NON-CONTINUOUS: EACH MESSAGE IS REPEATED AT THE LISTED RATE WHEN IT IS AVAILABLE, BUT MAY NOT BE AVAILABLE AT ALL TIMES. EXAMPLES ARE MODE-ORIENTED DATA AND TEST DATA.

2.1.4. BUS OPERATIONS (CONTINUED)

- (3) "BURST" TRANSMISSIONS: THESE MESSAGES ARE INTENDED TO INITIATE A SPECIFIC ACTION SUCH AS RADIO TUNING OR TO ANNOUNCE A SPECIFIC RESULT SUCH AS CAPTURE POINTS, ETC. EACH "NEW" MESSAGE REQUIRES A COMPLETE BURST TO BE SENT, AND THE BITS DEFINING THE ORIGINAL ACTION MAY NOT CHANGE STATE DURING THE BURST. TWO "BURSTS" MAY OVERLAP, BUT EACH ACTION IS REQUIRED TO HAVE 16 REPETITIONS SENT ONCE IN EACH OF 16 SUCCESSIVE FRAMES AT THE BUS UPDATE RATE OF 20/SECOND TO ENSURE SUCCESSFUL RECEPTION, SYNCHRONIZATION, DECODING, AND DISPOSITION BY THE RECEIVER. ALSO NOTE THAT THE TERMINATION OF THE BURST IS AN ESSENTIAL FEATURE OF THIS OPERATION, AND THE BURST MAY NOT BE EXTENDED, NOR SHOULD IT BE REPEATED UNLESS A NEW MESSAGE IS AVAILABLE. IT IS EXPECTED THAT THE RECEIVING UNIT WILL ACCEPT AS A VALID MESSAGE THE FIRST MESSAGE BLOCK WHICH CONTAINS A VERIFIABLE CHECKSUM.

COMMENT: WHEN THE BURST MODE IS USED TO TRANSMIT THE ACTUATION OF A MOMENTARY-ACTION SWITCH SUCH AS A PUSHBUTTON, BOTH STATES OF THE BUTTON MUST BE TRANSMITTED AS SEPARATE BURSTS IN ORDER THAT THE RECEIVING DEVICE MAY RECOGNIZE THE RELEASE OF THE SWITCH WHICH MAY OR MAY NOT BE COINCIDENT WITH THE END OF THE FIRST BURST. CONTROL ACTIONS WHICH ARE INTENDED TO REMAIN IN EFFECT SO LONG AS THE SWITCH IS PRESSED MAY THUS BE COMMANDED REMOTELY. NOTE 2, ADDRESS 30 FOR EXAMPLE, REFERS TO THE REMOTE INITIATION OF THE ACTIVE/PRESET EXCHANGE FUNCTION OF THE VHF COMM RADIO, USING BIT 5 BYTE (1). A REMOTE BUTTON, WHEN PUSHED, WOULD INITIATE A "BURST" OF ADDRESS 30 WITH BIT 5 BYTE (1) SET TO A "1", SIMULATING THE ACTION OF THE SWITCH ON THE CONTROL PANEL. UPON RELEASE OF THE REMOTE BUTTON, A SECOND "BURST" OF ADDRESS 30 WOULD BE SENT WITH THE SAME BIT SET TO AN "0", SIMULATING RELEASE OF THE CONTROL PANEL SWITCH. NOTE THAT ADDRESS 30 CONTROLS FREQUENCY TUNING OF THE RADIO AS WELL, AND TO AVOID RE-TUNING ACTION AT THE SAME TIME, IT IS NECESSARY TO RESET THE "TUNE VALID", BIT 7 TO AN "0" FOR THE MOMENTARY ACTION BURSTS.

ADDITIONAL COMMENT: A BUS MAY CONTAIN ANY OR ALL OF THESE FORMS OF TRANSMISSIONS. NOTE THAT FORMAT (3), "BURST" COULD WELL BE THE ONLY TRAFFIC ON A GIVEN BUS AT A GIVEN MOMENT, AND IF THIS IS THE CASE, THE BURST TRANSMISSION MUST INCLUDE THE NECESSARY SYNCHRONIZATION BLOCKS AS DESCRIBED ABOVE, ONE FOR EACH MESSAGE FRAME IN THE BURST. REFER TO SYNCHRONIZATION PARAGRAPH BELOW AS WELL.

2.1.5. MODULATION

DATA IS TRANSMITTED IN THE NRZ (NON RETURN TO ZERO) FORMAT, WITH A LOGIC "0" STATE BEING LINE B POSITIVE WITH RESPECT TO LINE A, AND A LOGIC "1" STATE BEING LINE A POSITIVE WITH RESPECT TO LINE B.

START BITS ARE LOGIC "0" STATE, AND STOP BITS ARE LOGIC "1" STATE.

COMMENT: THE DATA LOGIC IS CHANGED BACK TO ORIGINAL DEFINITION. EDITION 7 CHANGED THE DATA LOGIC TO MATCH THE DEFINITION IN RS-422-A PARAGRAPH 4.1, WHICH IS NOW FOUND TO BE OPPOSITE FROM THAT IMPLEMENTED IN COLLINS EQUIPMENT USING C.S.D.B. INTERFACES.

2.1.6. BUS SPEEDS

THERE WILL BE ONLY (2) BUS SPEEDS IN USE, AS FOLLOWS:

LO-SPEED BUS:12,500 BITS/SECOND, $\pm 0.1\%$
HI-SPEED BUS:50,000 BITS/SECOND, $\pm 0.1\%$

2.2. ELECTRICAL STANDARDS

COMMENT: THERE IS NO INTENT TO MAKE THIS STANDARD MORE RESTRICTIVE THAN NECESSARY FOR PROPER OPERATION. AN AIRCRAFT AVIONICS ENVIRONMENT IS SEVERE IN MANY RESPECTS, HOWEVER, AND THESE CONDITIONS MUST BE ADDRESSED REALISTICALLY IF LARGE SYSTEMS ARE TO HAVE ANY HOPE OF CORRECT AND RELIABLE OPERATION.

2.2.1. VOLTAGE AND IMPEDANCE LEVELS

LINE DRIVER AND LINE RECEIVER DEVICES USED SHALL CONFORM TO THE LEVELS SPECIFIED IN RS-422A. **COMMENT:** THE INTENDED RECEIVER DESIGN IS THE ONE DEFINED AS A DIFFERENTIAL INPUT, HIGH IMPEDANCE CIRCUIT.

2.2.2. RISE AND FALL TIMES

RISE AND FALL TIMES ARE TO BE AS DEFINED IN RS-422A EXCEPT THAT THEY SHALL NOT BE LESS THAN 0.8 NOR MORE THAN 1.0 MICROSEC(HI-SPEED BUS) NOR MORE THAN 8.0 MICROSEC (LO-SPEED BUS).

2.2.3. CAPACITY LOADING

- A. EACH RECEIVER SHALL HAVE A LINE-TO-GROUND OR LINE-TO-LINE CAPACITY NOT TO EXCEED 600 PICO FARADS.
- B. EACH TRANSMITTER SHALL BE DESIGNED TO DRIVE A CAPACITY LOAD OF NOT LESS THAN 0 PICO FARADS NOR MORE THAN 12000 PICO FARADS.

COMMENT: THIS REQUIREMENT IS INTENDED TO PROVIDE PROPER OPERATION WITH RECEIVERS NUMBERING FROM 1 TO 10, AND CAPACITANCES DUE TO INSTALLATION RANGING FROM 0 TO 6000 PICO FARADS. INSTALLATIONS WITH FEWER THAN 10 RECEIVERS ON A GIVEN BUS MAY ALLOW AN ADDITIONAL 600 PICO FARADS FOR EACH RECEIVER LESS THAN 10.

2.2.4. FAULT PROTECTION

- A. RECEIVERS SHOULD BE PROTECTED AGAINST DAMAGE DUE TO "HOT SHORTS" TO EITHER 115 VAC OR TO +28 VDC. THE PROTECTION AGAINST 115 VAC SHORTS SHOULD BE VIEWED AS GOOD DESIGN PRACTICE IN ANY CASE, BUT MAY BE WAIVED AS A REQUIREMENT IF NEITHER THE SOURCE NOR THE RECEIVER LRU'S USE 115 VAC. IN THIS CASE, HOWEVER, THE PROTECTION AGAINST HOT SHORTS IN THE AIRCRAFT INTERCONNECT RESULTS FROM THE USE OF SHIELDED WIRE WITH THE SHIELD GROUNDED AT BOTH ENDS, AND A NOTE TO THAT EFFECT IS RECOMMENDED AS PART OF THE INSTALLATION MANUAL. SEE FIGURE 1 FOR RECOMMENDED GROUNDING INFORMATION.
- B. TRANSMITTERS WILL BE DIFFICULT OR IMPOSSIBLE TO PROTECT AGAINST HOT SHORT DAMAGE, BUT THE DESIGN SHOULD BE SUCH THAT THE RESULTING DAMAGE WILL NOT BE PROPAGATED TO OTHER CIRCUITS. NO FUNCTIONAL LOSS EXCEPT THE SHORTED BUS OUTPUT SHOULD OCCUR.

2.2.5. RECEIVER AND TRANSMITTER CIRCUITS

SEE FIGURE 2. "TYPICAL LINE RECEIVER"

SEE FIGURE 3. "TYPICAL LINE TRANSMITTER"

2.3. DEFINITIONS

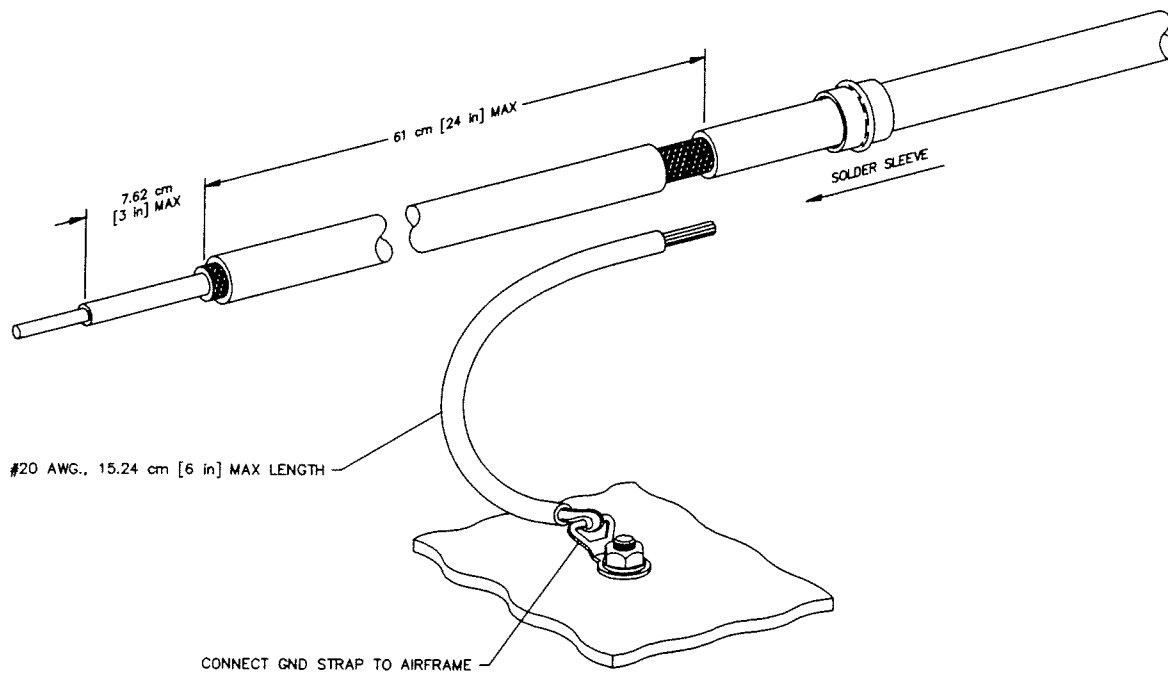
2.3.1. BYTE DEFINITION

DATA IS TRANSMITTED IN THE BYTE FORMAT USING ELEVEN (11) BIT GROUPS. EACH GROUP CONSISTS OF A START BIT, FOLLOWED BY 8 DATA BITS, FOLLOWED BY A PARITY BIT (ODD), AND ENDING WITH A STOP BIT. THE 8 DATA BITS (ONE BYTE) ARE NUMBERED 0 THRU 7 FOR EACH BYTE, AND ARE TRANSMITTED LSB FIRST. BYTES MAY CONTAIN ANY DATA WHATEVER EXCEPT FOR THE TWO CASES BELOW:

- (1) ADDRESS BYTE: MUST CONSIST OF PRE-ASSIGNED ADDRESS AS LISTED BELOW TO IDENTIFY THE DATA WHICH FOLLOWS IN THAT MESSAGE BLOCK.
- (2) STATUS BYTE: THIS BYTE, IF USED, CONTAINS INFORMATION ON THE SOURCE, MODE, AND VALIDITY OF THE DATA WHICH FOLLOWS IN THAT MESSAGE BLOCK.

2.3.2. MESSAGE BLOCK DEFINITION

A MESSAGE BLOCK IS DEFINED AS A SINGLE, SERIAL MESSAGE CONSISTING OF A FIXED NUMBER OF BYTES TRANSMITTED IN A FIXED SEQUENCE. THE MESSAGE IS IDENTIFIED BY A "LABEL", OR ADDRESS WHICH IS ALWAYS THE FIRST BYTE, BYTE (0). EACH MESSAGE MAY BE SUBDIVIDED IN ANY PREARRANGED WAY WHATEVER TO CONTAIN ONE OR MORE PARAMETERS. REFER TO SECTIONS BELOW FOR THE PREASSIGNED DEFINITIONS OF EACH MESSAGE. ON A GIVEN BUS, ALL MESSAGE BLOCKS MUST CONTAIN THE SAME NUMBER OF BYTES.

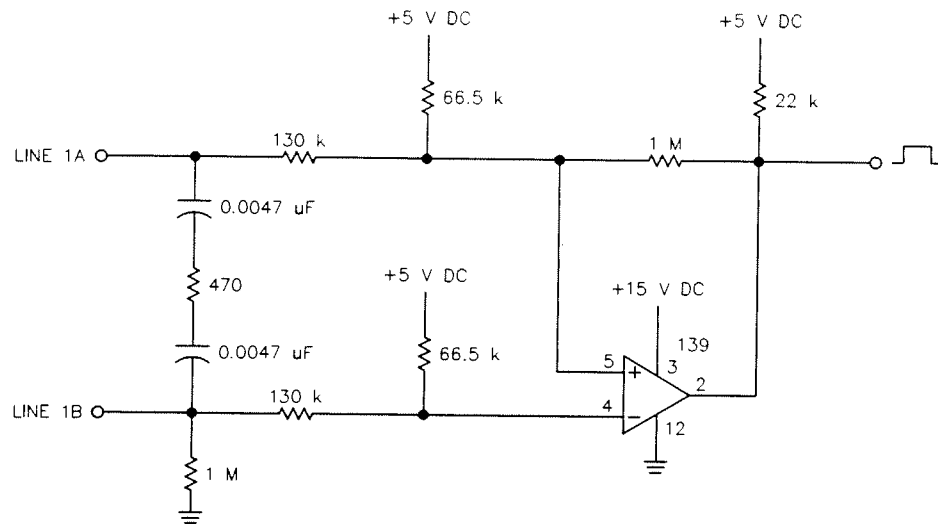


NOTES:

1. ALL DIMENSIONS ARE TO BE KEPT AS SHORT AS POSSIBLE. MAX LENGTHS MUST NOT BE EXCEEDED.
2. SHIELDED WIRES ARE TO BE CONNECTED TO AIRFRAME GROUND AT BOTH ENDS. THIS INCLUDES ALL SECTIONS OR "BREAKS" IN A MULTISECTION RUN.

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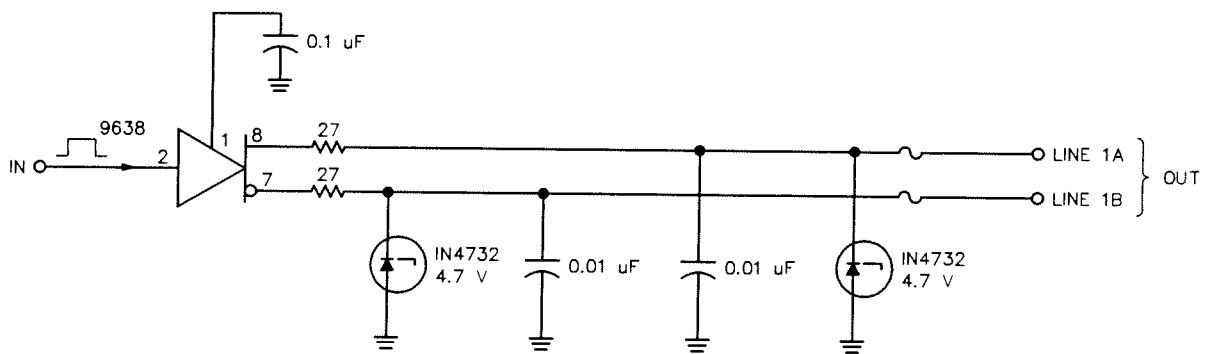
FIGURE 1.
GROUNDING OF AIRCRAFT INTERCONNECT SHIELDED WIRES



NOTE: RESISTANCE VALUES ARE IN OHMS.

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FIGURE 2
TYPICAL LINE RECEIVER



NOTE: RESISTANCE VALUES ARE IN OHMS.

CG0-4419-03-AC-3

FIGURE 3.
TYPICAL LINE TRANSMITTER

2.3.2. MESSAGE BLOCK DEFINITION (CONTINUED)

COMMENT: EACH MESSAGE IS INTENDED TO BE IDENTIFIED SOLELY BY ITS ADDRESS BYTE, AND NOT BY THE ORDER OR SEQUENCE OF TRANSMISSION. THE SEQUENCES AS LISTED BELOW FOR EACH BUS ARE ALPHABETICAL FOR EASE OF READING, AND DO NOT REPRESENT THE REAL TRANSMISSION ORDER. THE ONLY RESTRICTION ON THE TRANSMITTER DESIGN IS THAT WHATEVER SEQUENCE IS CHOSEN, IT MUST REMAIN FIXED FOR A PARTICULAR RADIO. NOTE THAT THIS DOES NOT PRECLUDE A NEW SEQUENCE AS A RESULT OF A SUBSEQUENT DESIGN CHANGE OR SERVICE BULLETIN. (SEE ALSO COMMENTS UNDER "REFRESH RATES".)

2.3.3. FRAME DEFINITION

A FRAME IS DEFINED AS THE INTERVAL FROM THE START OF A SYNC BLOCK TO THE START OF THE NEXT SYNC BLOCK.(SEE T1 ON FIGURE 4). A FRAME IS A FIXED INTERVAL OF TIME ON A GIVEN BUS, AND IS DETERMINED BY THE MAXIMUM UPDATE RATE REQUIRED ON THAT BUS. NOTE THAT A SINGLE FRAME MAY NOT CONTAIN ALL THE MESSAGES AVAILABLE ON A GIVEN BUS. (SEE REFRESH RATES BELOW.)

2.4. SYNCHRONIZATION

2.4.1. SYNC CHARACTER

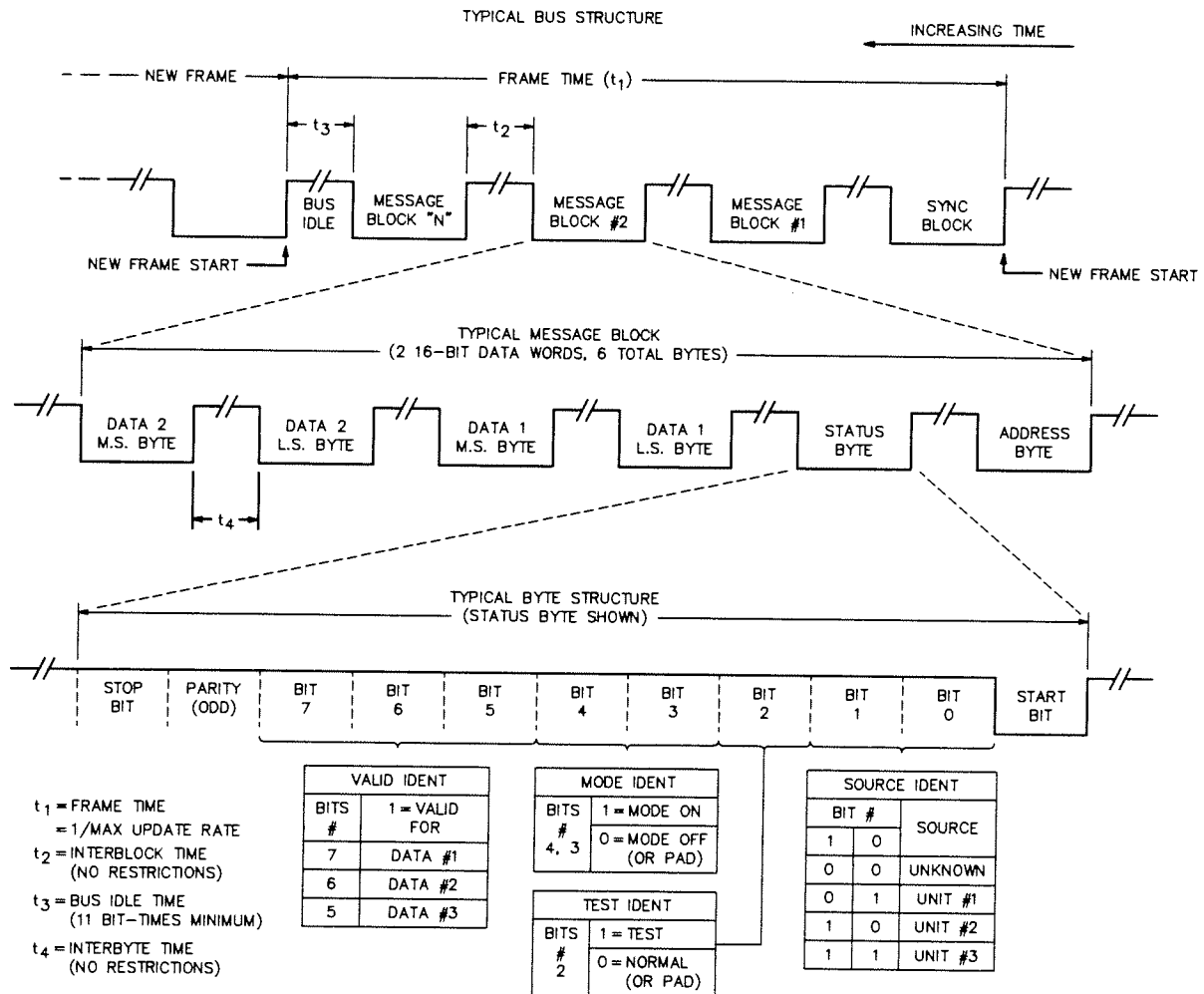
SYNCHRONIZATION OF EACH BYTE OR CHARACTER IS ACHIEVED VIA THE START AND STOP BITS DESCRIBED ABOVE. THE HEXADECIMAL CHARACTER "A5" IS THE DESIGNATED SYNC CHARACTER FOR USE IN ESTABLISHING MESSAGE BLOCK SYNC, BUT THIS DOES NOT PRECLUDE ITS USE AS VALID DATA (SEE BELOW):

2.4.2. SYNC BLOCK

A SYNC BLOCK CONSISTS OF "N" BYTES OF REPEATED SYNC CHARACTERS "A5", WHERE "N" IS THE NUMBER OF BYTES FIXED FOR THAT BUS. THE SYNC BLOCK MARKS THE BEGINNING OF A NEW FRAME. THIS BLOCK OF "N" BYTES IS GUARANTEED UNIQUENESS BECAUSE "A5" IS NEVER USED AS AN ADDRESS FOR ANY DATA MESSAGE BLOCKS WHICH FOLLOW. EVEN IF "A5" APPEARS IN A DATA PORTION OF A MESSAGE, IT COULD APPEAR N-1 TIMES AT MOST.

2.4.3. SYNC FREQUENCY

THE FREQUENCY OF SYNCHRONIZATION WAS SELECTED TO BE THE SAME AS THE MAXIMUM UPDATE RATE FOR EACH BUS IN ORDER THAT ESSENTIAL DATA MAY BE VALIDATED AND USED WITH MINIMUM DELAY, OR RECOVERED QUICKLY AFTER OUTAGES. (SYNCHRONIZATION LESS OFTEN WOULD SUFFICE FOR SOME DATA, BUT KEEPING A SINGLE STANDARD IS DESIRABLE, ESPECIALLY WHEN THE ESSENTIAL NATURE OF THE DATA IS NOT CLEAR.)



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FIGURE 4.
TYPICAL BUS STRUCTURE

2.5. DATA REFRESH RATES

DATA REFRESH RATES HAVE BEEN SELECTED TO TRANSFER ALL USEFUL INFORMATION CONTENT WHILE MAINTAINING AS LOW A BUS ACTIVITY AS POSSIBLE. THE COLUMN LABELED "UPDATE RATE" IN THE TABLES WHICH FOLLOW IDENTIFIES THE NOMINAL FREQUENCY, $\pm 10\%$, WITH WHICH EACH PARAMETER OR LABEL APPEARS ON A PARTICULAR BUS. IT IS INTENDED THAT EACH SUCH APPEARANCE BE THE RESULT OF A NEW COMPUTATION OR A NEW SAMPLING OF THE DATA. NEW COMPUTATIONS OR SAMPLINGS MAY TAKE PLACE MORE FREQUENTLY THAN THEY ARE TRANSMITTED, BUT SHOULD NOT TAKE PLACE LESS OFTEN. FROM THE BUS USER'S VIEW, THEN, THE TERMS UPDATE RATE AND REFRESH RATE ARE USED INTERCHANGEABLY AND ARE MEANT TO BE SYNONYMOUS. A FRAME AS DEFINED ABOVE WILL THUS HAVE THE SAME RATE AS THE MOST FREQUENT PARAMETER ON THAT BUS. ALL MESSAGES ON A GIVEN BUS NEED NOT HAVE THE SAME UPDATE RATE BY VIRTUE OF TRANSMITTING SOME MESSAGES LESS OFTEN THAN EVERY FRAME. THIS PROVISION IS INTENDED TO ALLOW FOR DATA WHICH IS AVAILABLE ONLY ON AN OCCASIONAL BASIS, AND NOT TO ALLOW VARIABLE UPDATE RATES. (SEE COMMENTS BELOW).

COMMENT: ONE CRITICAL ASPECT OF REFRESH RATES IS THAT A VARIABLE OR MODULATED RATE HAS THE POTENTIAL OF INTRODUCING "ALIASING" NOISE INTO THE SIGNAL PASSBAND. THE AMOUNT OF VARIABILITY TOLERABLE WITHOUT PROBLEMS IS VERY APPLICATION SENSITIVE, BUT DESIGNERS OF SOURCE LRU'S SHOULD ADDRESS THIS PROBLEM VERY CAREFULLY. ABSOLUTELY FIXED UPDATE RATES MAY NOT BE ACHIEVABLE ECONOMICALLY, BUT THE FOLLOWING IS A RECOMMENDED GUIDELINE:

SAMPLE-TO-SAMPLE TIMING: $\pm 1\%$ OF THE INTERVAL
VARIATION OVER 100 SAMPLES: $\pm 5\%$ OF THE INTERVAL

A SECOND SERIOUS ASPECT OF REFRESH RATES IS THAT THE NOMINAL RATES LISTED BELOW FOR EACH BUS REPRESENT FIRM DESIGNS OF CURRENT PRODUCTION UNITS, AND SIGNIFICANT DEPARTURE FROM THOSE NOMINALS BY NEW DESIGNS RUNS SERIOUS RISK THAT THE OUTPUTS WILL NO LONGER BE INTERCHANGEABLE. IT IS RECOMMENDED THAT IF A PROPOSED NEW UPDATE RATE EXCEEDS THE $\pm 10\%$ REQUIREMENT LISTED ABOVE, THEN ALL KNOWN APPLICATIONS FOR THAT BUS DATA BE CAREFULLY REVIEWED.

2.6. BUS TIMING

2.6.1. IDLE TIME & GAPS

THE IDLE STATE FOR ANY BUS IS A LOGIC "1", AND THE IDLE TIME MAY BE DISTRIBUTED WITHIN THE FRAME IN ANY DESIRED MANNER SO LONG AS THE FOLLOWING RESTRICTION IS OBSERVED: PRIOR TO TRANSMITTING THE SYNC BLOCK MESSAGE, THE BUS MUST BE IN THE IDLE STATE FOR A PERIOD NOT LESS THAN 11 BIT-TIMES. BOTH THE INTER-BLOCK TIME AND THE INTER-BYTE CAN BE AS SMALL AS ZERO.

NOTE: WHATEVER DISTRIBUTION IS PLANNED FOR THE BUS IDLE TIME, IT MUST NOT VARY IN A MANNER TO VIOLATE THE DATA REFRESH RATE RESTRICTION DESCRIBED ABOVE.

3. MESSAGE BLOCK FORMAT

3.1. BYTE SEQUENCE

THE BYTE SEQUENCE IS FIXED, AND MUST BE AS LISTED BELOW FOR EACH MESSAGE BLOCK. IN GENERAL, THE SEQUENCE IS AS FOLLOWS:

BYTE (0): ADDRESS BYTE
BYTE (1): STATUS BYTE OR FIRST DATA BYTE
BYTE (2): DATA BYTE
BYTES (3) THRU (N): DATA BYTES

3.2. PARAMETER GROUPING

EACH MESSAGE BLOCK CONTAINS DATA RELATING TO ONE OR MORE PARAMETERS IN ORDER TO REDUCE THE OVERHEAD ASSOCIATED WITH SERIAL BUS TRANSMISSION.

3.3. CONVENTIONS

3.3.1. ADDRESS BYTE

HEXADECIMAL ADDRESSES ARE READ FROM LEFT TO RIGHT, ASSUMING THAT BIT 0 IS WRITTEN AS THE RIGHT-MOST BIT. FOR EXAMPLE:

 BIT # 76543210
HEX ADDRESS 2E= 00101110
 / 2 / E /

3.3.2. STATUS BYTE

A STATUS BYTE IS DEFINED FOR SOME MESSAGE BLOCKS WHICH GIVES INFORMATION CONCERNING THE SOURCE, MODE, AND VALIDITY OF THE DATA CONTAINED IN THAT BLOCK. THE STATUS BYTE IS NOT USED IN ALL BLOCKS, BUT WHERE IT IS USED, THE FORMAT IS AS FOLLOWS:

BIT (0): RESERVED FOR SOURCE IDENTIFICATION
BIT (1): RESERVED FOR SOURCE IDENTIFICATION
BIT (2): TEST CONDITION IDENTIFIER
BIT (3): MODE CONDITION IDENTIFIER
BIT (4): MODE CONDITION IDENTIFIER
BIT (5): VALIDITY IDENTIFIER, DATA 3
BIT (6): VALIDITY IDENTIFIER, DATA 2
BIT (7): VALIDITY IDENTIFIER, DATA 1

NOTE: DATA BLOCKS CONTAINING ONLY 1 DATA PARAMETER WILL USE ONLY BIT 7 AS A VALID, WITH BITS 5 AND 6 BEING PADS.

3.3.3. DATA BYTE(S)

DATA BYTES MAY CONTAIN ANY DATA WHATEVER, WITH THE ONLY RESTRICTION THAT DATA WHICH REQUIRES MORE THAN ONE BYTE MUST BE TRANSMITTED LEAST SIGNIFICANT BYTE FIRST.

3.3.4. BIT LOGIC

BITS WITHIN EACH BYTE ARE DEFINED AS EITHER "ACTIVE" BITS OR "PAD" BITS. THE ACTIVE BITS ARE EITHER "1" OR "0" DEPENDING UPON THE MESSAGE, WHILE THE PAD BITS ARE ALWAYS "0", AND CONTAIN NO INFORMATION. FOR INFORMATION ONLY 1-BIT LONG (IE, VALID, TEST, ETC.) THE ACTIVE OR ACTUATED OR ON CONDITION IS REPRESENTED BY A "1", AND THE OPPOSITE BY AN "0". REFER TO THE SPECIFIC ASSIGNMENTS BELOW FOR MESSAGE DETAILS ON A PARTICULAR MESSAGE BLOCK.

3.3.5. BIT SEQUENCE

THE BITS WITHIN EACH BYTE ARE ALWAYS TRANSMITTED LEAST SIGNIFICANT BIT FIRST, AND ARE NUMBERED 0 THRU 7 BEGINNING WITH THE FIRST DATA BIT RECEIVED (LSB). (THE START, PARITY, AND STOP BITS ARE NOT NUMBERED.)

3.3.6. ACCURACY AND RESOLUTION

THE RESOLUTION CAPABILITY DEFINED IN THE DATA FORMATS BELOW DOES NOT MEAN THAT THE SOURCE HAS ACCURACY COMMENSURATE WITH THAT FORMAT OR EVEN THAT THE SOURCE HAS RESOLUTION TO THAT LEVEL. REFER TO THE EQUIPMENT SPECIFICATIONS FOR THAT SOURCE.

3.3.7. SIGNS

THE SIGN BIT FOR DIRECTIONAL QUANTITIES IS ENCODED:

POSITIVE FOR:UP, RIGHT, FORE, NORTH, EAST, AND TO
NEGATIVE FOR:DOWN, LEFT, AFT, SOUTH, WEST, AND BACK

THE DIRECTIONAL SENSE FOR ALL ACCELERATIONS, RATES, AND DEVIATIONS IS POSITIVE IN THE SAME DIRECTION AS FOR THE BASE QUANTITIES. THESE DIRECTIONS REFER TO AIRCRAFT MOTIONS AND THE RESPONSE OF SENSING DEVICES TO THAT MOTION.

THE SENSING FOR COMMANDS AND DEVICES USING OR GENERATING COMMANDS IS SUCH THAT COMMANDS ENCODED AS POSITIVE WILL CAUSE IN AIRCRAFT MOTION IN A POSITIVE DIRECTION AS DEFINED ABOVE. IT IS THE RESPONSIBILITY OF THE BUS RECEIVER WHO GENERATES COMMANDS FROM ONE OR MORE PIECES OF SENSED DATA TO COMBINE THEM IN THE CORRECT POLARITIES TO ACHIEVE PROPER CONTROL ACTION, AND TO ENCODE THE RESULTING COMMAND MESSAGE AS DEFINED ABOVE.

NOTE: ILS DEVIATION ON THE EFIS OUTPUT BUS ADDRESS C7 IS ENCODED AS A SENSING DEVICE, IE, + = FLY DOWN, FLY LEFT. ILS DEVIATION ON THE NAV BUS, ADDRESS 22 IS ENCODED AS A COMMAND DEVICE, IE, + = FLY UP, FLY RIGHT.

4. SYSTEM, BUS, ADDRESS, & DATA ASSIGNMENTS

4.1. BUS INDEX, ASSIGNED BUSES (LISTED ALPHABETICALLY)

COMMENT: THE BUS DESIGNATORS ARE TAKEN FROM THE TYPE NUMBER OF THE BOX WHICH CONTAINS THE TRANSMITTER. WHEN A BOX HAS MORE THAN ONE OUTPUT BUS, THEY ARE NUMBERED 1-N. OUTPUTS WHICH ARE SIMPLY A BUFFERED VERSION OF ANOTHER BUS ARE IDENTIFIED BY SUFFIXES A, B, ETC.

<u>BUS NAME</u>	<u>SYSTEM</u>	<u>PAGE #</u>
ADC BUS 1A,1B	AIR DATA SYSTEMS	18
ADF BUS 1	NAVIGATION (ADF)	22
ADF CTL BUS 1	NAVIGATION (ADF)	22
AHC BUS 1A,1B,1C	ATTITUDE/HEADING (AHRS)	17
ATC CTL BUS 1	ATC SYSTEM	18
ATC XPDR BUS 1	ATC SYSTEM	18
CDU BUS 1A,1B	NAVIGATION (LONG RANGE)	22
DME BUS 1A,1B	NAVIGATION (DME)	23
DPU BUS 1A (NOTE 1)	EFIS SYSTEM	20
FCC BUS 1 OR 2	FLIGHT CONTROL	21
FGC BUS 1	FLIGHT CONTROL	21
MCS BUS 1 (NOTE 2)	HEADING	17
MPU BUS 1 (NOTE 1)	EFIS SYSTEM	20
NAV BUS 1A,1B	NAVIGATION (VHF)	23
NAV CTL BUS 1A,1B	NAVIGATION (VHF)	23
VHF COMM BUS 1	COMMUNICATIONS (VHF)	19
VHF COMM CTL BUS 1A,1B	COMMUNICATIONS (VHF)	19
VNI BUS 1	NAVIGATION (VERTICAL)	23

NOTE 1: THE BUS NAME OF DPU AND MPU REFERS TO A "DISPLAY PROCESSOR UNIT" AND "MULTIFUNCTION PROCESSOR UNIT" OF AN EFIS SYSTEM THAT HAS THE PROCESSOR SEPARATE FROM THE EFIS DISPLAY. FOR SYSTEMS WITH SELF-CONTAINED EFIS DISPLAY UNITS (NO EXTERNAL DRIVE UNITS), THE OUTPUT BUS FROM THE DISPLAY SHOULD BE EQUIVALENT TO THE DPU AND/OR MPU BUS DESCRIBED IN THIS DOCUMENT IN ORDER TO COMPLY WITH THIS STANDARD.

NOTE 2: MCS BUS REFERS TO A MAGNETIC COMPASS SYSTEM BUS OUTPUT THAT CONTAINS MAGNETIC HEADING INFORMATION.

4.2. ADDRESS ORDER INDEX, ASSIGNED ADDRESSES

TITLE: ADDRESS ORDER INDEX

ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION	PAGE #	SUB- SYSTEM
00	NOT USED--ILLEGAL ADDRESS FOR DATA--		
01	RESERVED FOR EXTENDED ADDRESSING		
02 THRU 0F			UNASSIGNED
10	VHF COMM FREQ	24	VHF COMM
11	VHF COMM DATA	25	VHF COMM
12	VHF COMM FREQ (8.33 KHZ CHAN)	26	VHF COMM
13	VHF COMM DATA (8.33 KHZ CHAN)	27	VHF COMM
14 THRU 15			UNASSIGNED
16 THRU 19	RESERVED	-	HF
1A THRU 1D			UNASSIGNED
1E	ATC CODE/ALTITUDE	28	ATC
1F	XPDR OUTPUT DATA	29	ATC
20	VHF NAV FREQ	30	VHF NAV
21	VOR DATA	31	VHF NAV
22	ILS DATA	32	VHF NAV
23	RESERVED	-	
24	DME FREQ	33	DME
25	DME FREQ & DIST	34	DME
26	DME TTS & VEL	35	DME
27	DME IDENT	36	DME
28 THRU 29		-	RESERVED
2A	STEER COMMANDS/ALT REF	37	FCS
2B	ADF DATA	38	ADF
2C	ADF FREQUENCY	39	ADF
2D THRU 2F		-	RESERVED

TITLE: ADDRESS ORDER INDEX (CONTINUED)

ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION	PAGE #	SUB- SYSTEM
30	REMOTE NAV TUNE	40	FMS
31	REMOTE COMM TUNE	41	FMS
32	REMOTE ADF TUNE	42	FMS
33	REMOTE ATC CODE	43	FMS
34	REMOTE DME TUNE	44	FMS
35	REMOTE COMM TUNE (8.33 KHZ)	46	FMS
36		-	RESERVED
37	REMOTE MLS CHANNEL AND GLIDE PATH TUNE	47	MLS
38	REMOTE MLS AZIMUTH AND BACK AZIMUTH TUNE	48	MLS
39	REMOTE HF TUNE, XMIT (RSVD)	-	FMS
3A	REMOTE HF TUNE, RECV (RSVD)	-	FMS
3B THRU 3F		-	RESERVED
40	RATES (BODY AXIS)	49	AHRS
41	ACCELERATIONS (BODY AXIS)	50	AHRS
42	ACCELERATIONS	51	AHRS
43	ATTITUDE/HEADING (EULER)	52	AHRS
44 THRU 45	RESERVED	-	AHRS
46	HEADING, MAGNETIC	53	MCS
47	AHRS CONTROL	54	AHRS
48	RESERVED	-	INERTIAL
49 THRU 4F		-	RESERVED
50 THRU 6F		-	UNASSIGNED
70	CRT ASCII MESSAGES	55	FCS
71	RESERVED	-	
72	VS/IAS/MACH REFERENCES	56	FCS
73	DELTA TORQUE REQUEST AND PWT STATUS	57	FCS
74 THRU 9F		-	UNASSIGNED
A0	PRESSURE/BARO ALTITUDES	58	ADS
A1	VERTICAL SPEED/IAS	59	ADS
A2	TAS/ISA DELTA TEMP	60	ADS
A3	MACH	61	ADS
A4	VS/IAS BUGS	62	ADS
A6	SAT, V_{MO}/M_{MO}	63	ADS
A7	TAT/PRESEL ALTITUDE	64	ADS
A8	M_{MO} /BARO SET	65	ADS
A9	RESERVED (ALT REPORTING)	-	ADS

TITLE: ADDRESS ORDER INDEX (CONTINUED)

ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION	PAGE #	SUB- SYSTEM
AA	FCS ANNUNCIATIONS	66	FCS
AB	FCS MODES	67	FCS
AC	SECONDARY IAS/MACH REFER- ENCE MARKERS	68	ADS
AD	LOW SPEED CUE AND SECONDARY IAS/MACH RANGE MARKERS	69	ADS
AF	AIR DATA CONTROL & STATUS	70	ADS
B0	PITCH CMD/VERTICAL DEV	71	VNAV/EFIS
B1	DISTANCE TO TRK/ALT	72	VNAV
B2	SEL ANGLE/COMP ANGLE	73	VNAV
B3	AIMPOINT ALT/VS SELECT	74	VNAV
B4 THRU BF		-	RESERVED
C0	EFIS CONTROL & STATUS	75	EFIS
C1	HDG/HDG ERROR	77	EFIS
C2	CRS ERROR (ACTIVE/NEXT)	78	EFIS
C3	RADIO ALT/DH	79	EFIS
C4	FCS LAT/VERT DEV (ACTIVE)	80	EFIS
C5	FCS LAT/VERT DEV (NEXT)	81	EFIS
C6	NAV ROLL/PITCH CMDS (ACTIVE)	82	EFIS
C7	LOCALIZER/GLIDESLOPE DEV	83	EFIS
C8	FCS DISTANCE (ACTIVE/NEXT)	84	EFIS
C9	WIND VEL/DIRECTION	85	EFIS
CA	SELECTED COURSE (ACTIVE/NEXT)	86	EFIS
CB	EFIS JOYSTICK CURSOR POSITION	87	EFIS
CC	EFIS KEYSWITCH STATUS	88	EFIS
CD	HDG/HDG ERROR	89	EFIS
CE	MIN & MAX STALL SPEEDS	90	EFIS
CF	RESERVED	-	EFIS
D0 THRU EF		-	UNASSIGNED
F0	RSVD FLT RECORDER TEST	-	---
F1	RSVD UNIVERSAL TEST INPUT	-	---
F2	RSVD UNIVERSAL TEST OUTPUT	-	---
F3	UNIVERSAL DIAGNOSTICS	91	---
F4	RESERVED	-	---
F5	UNIVERSAL ASCII PAGE DATA	93	---
F6 THRU FE			UNASSIGNED
FF	NOT USED--ILLEGAL ADDRESS FOR DATA--		

4.3. BUS CHARACTERISTICS (ALPHABETICAL BY SYSTEM)

4.3.1. ATTITUDE AND HEADING SYSTEMS BUS CHARACTERISTICS

KILOBITS/ SECOND	BYTES/ BLOCK	FRAMES/ SECOND	BLOCKS/ FRAME (MAX)
50	8	52	10
ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION		UPDATES/ SECOND
AHC BUS 1A/1B/1C (AHRS SYSTEM)			
A5	SYNC		52
40	RATES (BODY AXIS)		52
41	ACCELERATIONS (BODY AXIS)		52
42	ACCELERATIONS		5.2
43	ATTITUDE/HEADING		52
44	RESERVED		52
45	RESERVED		52
47	AHRS CONTROL		5.2
F3	AHRS DIAGNOSTICS		5.2
F5	AHRS ASCII PAGE DATA		5.2
3 BLOCK TIMES/FRAME UNASSIGNED			

■ MCS BUS 1 (MCS COMPASS SYSTEM)

KILOBITS/ SECOND	BYTES/ BLOCK	FRAMES/ SECOND	BLOCKS/ FRAME (MAX)
12.5	6	20	9
ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION		UPDATES/ SECOND
A5	SYNC		20
46	MAGNETIC HEADING		20
7 BLOCK TIMES/FRAME UNASSIGNED			

4.3.2. AIR DATA SYSTEM BUS CHARACTERISTICS

KILOBITS/ SECOND	BYTES/ BLOCK	FRAMES/ SECOND	BLOCKS/ FRAME (MAX)
---------------------	-----------------	-------------------	------------------------

12.5	6	20	9
------	---	----	---

ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION	UPDATES/ SECOND
------------------	------------------------------	--------------------

ADC BUS1A/1B/1C (AIR DATA SYSTEM)

A5	SYNC	20
A0	PRESSURE/BARO ALTITUDES	20
A1	VERT SPEED/IAS	20
A2	TAS/ISA DELTA TEMP	20
A3	MACH	10
A4	VS/IAS BUGS	10
A6	SAT, V_{MO}/M_{MO}	10
A7	TAT/PRESEL ALTITUDE	10
A8	M_{MO} /BARO SET	10
A9	RESERVED (ALT REPORTING)	10
AC	SECONDARY IAS/MACH REF MKRS	10
AD	LOW SPEED CUE AND SECON- DARY IAS/MACH RANGE MKRS	10
F3	ADC DIAGNOSTICS	10
F5	ADC ASCII PAGE DATA	10

0 BLOCK TIMES/FRAME UNASSIGNED

ADC INPUT CONTROL (AIR DATA SYSTEM)

AF	AIR DATA CONTROL & STATUS	10
----	---------------------------	----

TBA BLOCK TIMES/FRAME UNASSIGNED

4.3.3. ATC SYSTEM BUS CHARACTERISTICS

KILOBITS/ SECOND	BYTES/ BLOCK	FRAMES/ SECOND	BLOCKS/ FRAME (MAX)
---------------------	-----------------	-------------------	------------------------

12.5	6	10	18
------	---	----	----

ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION	UPDATES/ SECOND
------------------	------------------------------	--------------------

ATC CTL BUS 1A/1B

A5	SYNC	10
1E	XPDR ATC CODE/ALTITUDE	10

16 BLOCK TIMES/FRAME UNASSIGNED

ATC TRANSPONDER BUS 1

A5	SYNC	10
1F	XPDR OUTPUT DATA	10
F3	XPDR DIAGNOSTICS	10

15 BLOCK TIMES/FRAME UNASSIGNED

4.3.4. COMMUNICATIONS SYSTEMS BUS CHARACTERISTICS

KILOBITS/ SECOND	BYTES/ BLOCK	FRAMES/ SECOND	BLOCKS/ FRAME (MAX)
12.5	6	10	18
ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION	UPDATES/ SECOND	
VHF COMM CTL BUS 1A/1B (VHF COMM SYSTEM)			
A5	SYNC	10	
10	VHF COMM FREQ	10	
12	VHF COMM FREQ (8.33 KHZ)	10	
15 BLOCK TIMES/FRAME UNASSIGNED			
VHF COMM BUS 1 (VHF COMM SYSTEM)			
A5	SYNC	10	
11	VHF COMM DATA	10	
13	VHF COMM DATA (8.33 KHZ)	10	
F3	VHF DIAGNOSTICS	10	
14 BLOCK TIMES/FRAME UNASSIGNED			

4.3.5. ELECTRONIC FLIGHT INSTRUMENT SYSTEMS BUS CHARACTERISTICS

KILOBITS/ SECOND	BYTES/ BLOCK	FRAMES/ SECOND	BLOCKS/ FRAME (MAX)
---------------------	-----------------	-------------------	------------------------

12.5	6	10	18
------	---	----	----

ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION	UPDATES/ SECOND
------------------	------------------------------	--------------------

DPU BUS1A (EFIS SYSTEM)

A5	SYNC	10
C0	EFIS CONTROL & STATUS	10
C1	HEADING/HDG ERROR	10
C2	CRS ERROR (ACTV/NEXT)	10
C3	RADIO ALTITUDE/DH	10
C4	FCS LAT/VERT DEV (ACTV)	10
C5	FCS LAT/VERT DEV (NEXT)	10
C6	FCS ROLL/PITCH CMD (ACTV)	10
C7	LOCALIZER/GLIDESLOPE DEV	10
C8	FCS DISTANCE (ACTV/NEXT)	10
C9	WIND VELOCITY/DIRECTION	10
CA	SELECT CRS (ACTV/NEXT)	10
CE	MIN AND MAX STALL SPEEDS	10
B0 (NOTE 1)	PITCH CMD/VERT DEV	10
F3	DPU DIAGNOSTICS	10

3 BLOCK TIMES/FRAME UNASSIGNED

NOTE 1: RE-TRANSMITTED FROM VNI BUS OR FROM LONG RANGE NAVIGATOR

MPU BUS 1 (EFIS SYSTEM)

A5	SYNC	10
CB (NOTE 2)	JOYSTICK CURSOR POSITION	10
CC	EFIS KEYSWITCH STATUS	10
CD	SELECT HDG (BUG) POSITION	10
A7 (NOTE 3)	TAT/PRE-ALT	10
F3	MPU DIAGNOSTICS	10

12 BLOCK TIMES/FRAME UNASSIGNED

NOTE 2: LABEL CB IS TRANSMITTED 20 TIMES AT EACH PRESS OF THE ENTER KEY ON THE MFD.

NOTE 3: RE-TRANSMITTED FROM ADC BUS

4.3.6. FLIGHT CONTROL SYSTEMS BUS CHARACTERISTICS

KILOBITS/ SECOND	BYTES/ BLOCK	FRAMES/ SECOND	BLOCKS/ FRAME (MAX)
12.5	8	10	14
ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION	UPDATES/ SECOND	

FCC BUS 1 OR 2 (FLIGHT CONTROL SYSTEM)

A5	SYNC	10
AA	FCS ANNUNCIATIONS	10
AB	FCS MODES	10
2A	STEER CMDs/ALT REF	10
70 (NOTE 1)	CRT ASCII MESSAGES (8 MESSAGES/FRAME)	10
72	VS/IAS/MACH REFERENCES	10
73	DELTA TORQUE REQUEST AND PWT STATUS	10
F3	FCC DIAGNOSTICS	0 OR 10
FF (NOTE 2)	NO USEFUL DATA	0 OR 10

0 BLOCK TIMES/FRAME UNASSIGNED

NOTE 1: 8 ASCII MESSAGES ALL ARE SENT WITH ADDRESS 70; ADDRESS 70 GOES OUT TO UNIQUE LOCATIONS ON CRT SCREEN. SEE DETAILED DEFINITIONS BELOW.

NOTE 2: ALL TIME SLOTS WILL BE FILLED IN ALL FRAMES WITH EITHER DATA OR FF BLOCKS.

FGC BUS 1 (FLIGHT CONTROL SYSTEM - SEE NOTE 3)

A5	SYNC	10
AA	FCS ANNUNCIATIONS	10
AB	FCS MODES	10

11 BLOCK TIMES/FRAME UNASSIGNED

NOTE 3: FOR SYSTEMS THAT ONLY COMMUNICATE MODE SELECTION AND ANNUNCIATION.

4.3.7. NAVIGATION SYSTEMS BUS CHARACTERISTICS

KILOBITS/ SECOND	BYTES/ BLOCK	FRAMES/ SECOND	BLOCKS/ FRAME (MAX)
12.5	6	10	18
ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION		UPDATES/ SECOND

ADF BUS 1A/1B (ADF SYSTEM)

A5	SYNC	10
2B	ADF DATA	10
F3	ADF DIAGNOSTICS	10

15 BLOCK TIMES/FRAME UNASSIGNED

ADF CTL BUS 1 (ADF SYSTEM)

A5	SYNC	10
2C	ADF FREQUENCY	10

16 BLOCK TIMES/FRAME UNASSIGNED

CDU BUS 1A,1B (FMS OR LONG RANGE NAV SYSTEM)

A5	SYNC	20
30 (NOTE 1)	REMOTE NAV TUNE	0 OR 20
31 (NOTE 1)	REMOTE COMM TUNE	0 OR 20
32 (NOTE 1)	REMOTE ADF TUNE	0 OR 20
33 (NOTE 1)	REMOTE ATC CODE	0 OR 20
34 (NOTE 1)	REMOTE DME TUNE	0 OR 20
39 (NOTE 1)	REMOTE HF TUNE, XMIT RSVD	0 OR 20
3A (NOTE 1)	REMOTE HF TUNE, RECV RSVD	0 OR 20
F3	CDU DIAGNOSTICS	0 OR 20

8 BLOCK TIMES/FRAME AVAILABLE (NOTE 2)

NOTE 1: THESE WORDS ARE BURSTS ONLY. SEE BUS OPERATIONS PARAGRAPH ABOVE.

NOTE 2: THIS BUS IS INTENDED TO BE BASICALLY A "BURST" BUS FOR REMOTE TUNING AND HAS AVAILABLE 8 BLOCK TIMES ANY OF WHICH MAY BE USED FOR ANY "BURST" MESSAGE, ALLOWING AS MANY AS 8 SIMULTANEOUS ACTIONS.

4.3.7. NAVIGATION SYSTEMS BUS CHARACTERISTICS (CONTINUED)

KILOBITS/ SECOND	BYTES/ BLOCK	FRAMES/ SECOND	BLOCKS/ FRAME (MAX)
12.5	6	10	18
ADDRESS (HEX)	MESSAGE BLOCK DESCRIPTION	UPDATES/ SECOND	
DME BUS 1A/1B (DME SYSTEM)			
A5	SYNC		10
25	DME FREQ & DIST (CH1)		10
25	DME FREQ & DIST (CH2)		10
25	DME FREQ & DIST (CH3)		10
26	DME TTS & VEL (CH1)		5
26	DME TTS & VEL (CH2)		5
26	DME TTS & VEL (CH3)		5
27	DME IDENT (CH1)		5
27	DME IDENT (CH2)		5
27	DME IDENT (CH3)		5
F3	DME DIAGNOSTICS		5
F5	DME ASCII PAGE DATA		5
10 BLOCK TIMES/FRAME UNASSIGNED			
NAV CTL BUS 1A/1B (VHF NAV SYSTEM)			
A5	SYNC		10
20	NAV FREQ		10
24	DME FREQ (CH 1)		10
24	DME FREQ (CH 2)		10
24	DME FREQ (CH 3)		10
13 BLOCK TIMES/FRAME UNASSIGNED			
NAV BUS 1A/1B (VHF NAV SYSTEM)			
A5	SYNC		10
21	VOR DATA		10
22	ILS DATA		10
F3	NAV DIAGNOSTICS		10
F5	VIR ASCII PAGE DATA		10
14 BLOCK TIMES/FRAME UNASSIGNED			
VNI BUS 1 (VERTICAL NAVIGATION INSTRUMENT)			
A5	SYNC		10
B0	PITCH CMD/VERTICAL DEV		10
B1	DISTANCE TO TRK/ALT		10
B2	SELECT ANGLE/COMP ANGLE		10
B3	AIMPOINT ALT/VS SELECT		10
13 BLOCK TIMES/FRAME UNASSIGNED			

5. DETAILED MESSAGE BLOCK DEFINITIONS (IN ADDRESS ORDER)

TITLE: VHF COMM FREQ								
ADDRESS 10								
BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0

(0)	[0 0 0 1]				[0 0 0 0]			
	[-----HEX 1-----]				[-----HEX 0-----]			
	[-----ADDRESS BYTE-----]							
(1)	FREQ VALID 1=VAL	PAD	XFR CTL 1=XFR	SQ LCH CTL (NOTE 2)	SI (NOTE 1)	TEST CTL 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	[8 4 2 1]				[PAD PAD PAD PAD]			
	[-----BCD-----]				[-----]			
	[-----0.001 MHZ-----]				[-----]			
(3)	[8 4 2 1]				[8 4 2 1]			
	[-----BCD-----]				[-----BCD-----]			
	[-----0.1 MHZ-----]				[-----0.01 MHZ-----]			
(4)	[PAD 4 2 1]				[8 4 2 1]			
	[-----BCD-----]				[-----BCD-----]			
	[-----10.0 MHZ-----]				[-----1.0 MHZ-----]			
(5)	[PAD PAD PAD PAD]				[PAD PAD PAD PAD]			
	[-----]				[-----]			
	[-----]							

NOTE 1: SOURCE IDENT BITS (3,1,0) BYTE (1)

3 1 0		3 1 0
0 0 0	ALL UNITS	1 0 0 ---
0 0 1	#1 UNIT	1 0 1 #1 PRESET
0 1 0	#2 UNIT	1 1 0 #2 PRESET
0 1 1	#3 UNIT	1 1 1 #3 PRESET

NOTE 2: SQUELCH CONTROL, BIT(4) BYTE (1)

0=AUTOMATIC SQUELCH
1=AUDIO ENABLED (SQUELCH DISABLED)

ADDRESS 11

NOTE 1: SOURCE IDENT BITS (3,1,0) BYTE (1)

NOTE 2: FREQ LIMIT, BITS (6,5) BYTE (1)

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TITLE: VHF COMM FREQ (8.33 KHZ CHANNELS)

ADDRESS 12

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
	-----				-----			
(0)	[0	0	0	1]	[0	0	1	0]
	[-----HEX 1-----]				[-----HEX 2-----]			
	[-----ADDRESS BYTE-----]							
(1)	FREQ	PAD	XFR	SQLCH	SI	TEST	SI	SI
	VALID		CTL	CTL		CTL	SOURCE	
	1=VAL		1=XFR	(NOTE 2)		1=ON	IDENT	
					(NOTE 1)		(NOTE 1)	
(2)	[8	4	2	1]	[PAD	PAD	PAD	PAD]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.001 MHZ-----]				[-----BCD-----]			
(3)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.1 MHZ-----]				[-----0.01 MHZ-----]			
(4)	[PAD	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----10.0 MHZ-----]				[-----1.0 MHZ-----]			
(5)	[PAD	PAD	PAD	PAD]	[PAD	PAD	PAD	PAD]
	[-----]				[-----]			
	[-----]				[-----]			

NOTE 1: SOURCE IDENT BITS (3,1,0) BYTE (1)

$\frac{3}{0}$	$\frac{1}{0}$	$\frac{0}{0}$	ALL UNITS	$\frac{3}{1}$	$\frac{1}{0}$	$\frac{0}{0}$	---
0	0	1	#1 UNIT	1	0	1	#1 PRESET
0	1	0	#2 UNIT	1	1	0	#2 PRESET
0	1	1	#3 UNIT	1	1	1	#3 PRESET

NOTE 2: SQUELCH CONTROL, BIT(4) BYTE (1)

0=AUTOMATIC SQUELCH
1=AUDIO ENABLED (SQUELCH DISABLED)

TITLE: VHF COMM DATA (8.33 KHZ CHANNELS)

ADDRESS 13

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
	-----	-----	-----	-----	-----	-----	-----	-----
(0)	[0	0	0	1]	[0	0	1	1]
	[-----HEX 1-----]				[-----HEX 3-----]			
	[-----ADDRESS BYTE-----]							
(1)	FREQ VALID 1=VAL	FREQ LIMIT B	FREQ LIMIT A	XMIT IND 1=ON	SI (NOTE 1)	SELF TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
	(NOTE 2)				(NOTE 1)		(NOTE 1)	
(2)	[8	4	2	1]	[PAD	PAD	PAD	PAD]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.001 MHZ-----]				[-----BCD-----]			
(3)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.1 MHZ-----]				[-----0.01 MHZ-----]			
(4)	[PAD	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----10.0 MHZ-----]				[-----1.0 MHZ-----]			
(5)	[PAD	PAD	PAD	PAD]	[PAD	PAD	PAD	PAD]
	[-----BCD-----]				[-----BCD-----]			
	[-----10.0 MHZ-----]				[-----1.0 MHZ-----]			

NOTE 1: SOURCE IDENT BITS (3,1,0) BYTE (1)

$\frac{3}{0}$	$\frac{1}{0}$	$\frac{0}{0}$	ALL UNITS	$\frac{3}{1}$	$\frac{1}{0}$	$\frac{0}{0}$	---
0	0	1	#1 UNIT	1	0	1	#1 PRESET
0	1	0	#2 UNIT	1	1	0	#2 PRESET
0	1	1	#3 UNIT	1	1	1	#3 PRESET

NOTE 2: FREQ LIMIT, BITS (6,5) BYTE (1)

$\frac{6}{0}$	$\frac{5}{0}$	NOT USED
0	1	135.975 MHZ
1	0	136.975 MHZ
1	1	151.975 MHZ

TITLE: TRANSPONDER ATC CODE/ALTITUDE

ADDRESS 1E

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	0	1]	[1	1	1	0]
	[-----HEX 1-----]				[-----HEX E-----]			
	[-----ADDRESS BYTE-----]							
(1)	CODE VALID 1=VAL	PAD	MODE 1=STBY	ALT REPR 1=ON	ATC IDENT 1=ON	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	[PAD	4	2	1]	[PAD	4	2	1]
	[-----ATC CODE C-----]				[-----ATC CODE D-----]			
	[-----BCD, (0-7 ONLY)-----]							
(3)	[PAD	4	2	1]	[PAD	4	2	1]
	[-----ATC CODE A-----]				[-----ATC CODE B-----]			
	[-----BCD, (0-7 ONLY)-----]							
(4)	[PAD	4	2	1]	[PAD	4	2	1]
	[-----ALT CODE C-----]				[-----ALT CODE D-----]			
	[-----ALTITUDE REPORTING, GRAY CODE-----]							
(5)	[PAD	4	2	1]	[PAD	4	2	1]
	[-----ALT CODE A-----]				[-----ALT CODE B-----]			
	[-----ALTITUDE REPORTING, GRAY CODE-----]							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	NOT USED

NOTE 2: CODE VALID APPLIES TO THE ATC CODE FIELD ONLY. THE ALTITUDE REPORTING, GRAY CODE FIELD (BYTES 4 AND 5) IS AN UNMONITORED PARAMETER. IN SOME SYSTEMS, ALL 0'S OR ALL 1'S IN THIS FIELD INDICATE INVALID DATA.

TITLE: TRANSPONDER OUTPUT DATA

ADDRESS 1F

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	0	1]	[1	1	1	1]
	[-----HEX 1-----]				[-----HEX F-----]			
	[-----ADDRESS BYTE-----]							
(1)	CODE VALID 1=VAL	XMIT 1=ON	MODE 1=STBY	ALT REPRT 1=ON	ATC IDENT 1=ON	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	[PAD	4	2	1]	[PAD	4	2	1]
	[-----ATC CODE C-----]				[-----ATC CODE D-----]			
	[-----BCD, (0-7 ONLY)-----]							
(3)	[PAD	4	2	1]	[PAD	4	2	1]
	[-----ATC CODE A-----]				[-----ATC CODE B-----]			
	[-----BCD, (0-7 ONLY)-----]							
(4)	[PAD	4	2	1]	[PAD	4	2	1]
	[-----ALT CODE C-----]				[-----ALT CODE D-----]			
	[-----ALTITUDE REPORTING, GRAY CODE-----]							
(5)	[PAD	4	2	1]	[PAD	4	2	1]
	[-----ALT CODE A-----]				[-----ALT CODE B-----]			
	[-----ALTITUDE REPORTING, GRAY CODE-----]							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	NOT USED

NOTE 2: CODE VALID APPLIES TO THE ATC CODE FIELD ONLY. THE ALTITUDE REPORTING, GRAY CODE FIELD (BYTES 4 AND 5) IS AN UNMONITORED PARAMETER. IN SOME SYSTEMS, ALL 0'S OR ALL 1'S IN THIS FIELD INDICATE INVALID DATA.

TITLE: VHF NAV FREQ

ADDRESS 20

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	0]	[0	0	0	0]
	[-----HEX 2-----]				[-----HEX 0-----]			
	[-----ADDRESS BYTE-----]							
(1)	FREQ VALID 1=VAL	PAD	PAD	MKR SENS 1=LOW	DME HOLD 1=ON	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.1 MHZ-----]				[-----0.01 MHZ-----]			
(3)	[PAD	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----10.0 MHZ-----]				[-----1.0 MHZ-----]			
(4)	[PAD	PAD	PAD	PAD]	[PAD	PAD	PAD	PAD]
	[-----]				[-----]			
	[-----]				[-----]			
(5)	[PAD	PAD	PAD	PAD]	[PAD	PAD	PAD	PAD]
	[-----]				[-----]			
	[-----]				[-----]			

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

TITLE: VOR DATA

ADDRESS 21

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	0]	[0	0	0	1]
	[-----HEX 2-----]				[-----HEX 1-----]			
	[-----ADDRESS BYTE-----]							
(1)	VOR VALID 1=VAL	FREQ VALID 1=VAL	2/5 TUNE 1=ENAB	MKR SENS 1=LOW	ROT MOD FILTER 1=ACTV	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	←-----LSB]				PAD	MARKER BEACONS		
	←-----]					INN	MDL	OUT
	←-----]					1=ON	1=ON	1=ON
	(NOTE 2)							
(3)	[MSB-----2'S COMPLEMENT, 12-BIT-----←							
	[-180-----DEGREES-----←							
	[-----VOR BEARING, FROM A/C TO STATION-----←							
(4)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.1 MHZ-----]				[-----0.01 MHZ-----]			
(5)	[PAD	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----10.0 MHZ-----]				[-----1.0 MHZ-----]			

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

NOTE 2: MARKER BEACON BITS (2,1,0) BYTE (2)

THESE 1-BIT MESSAGES ARE TRANSMITTED UPON MARKER RECEPTION
INDEPENDENTLY OF THE STATE OF THE FLAG BITS (7,6) BYTE (1)

TITLE: ILS DATA

ADDRESS 22

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	0]	[0	0	1	0]
	[-----HEX 2-----]				[-----HEX 2-----]			
	[-----ADDRESS BYTE-----]							
(1)	GS VALID 1=VAL	LOC VALID 1=VAL	D'LYD ILS 1=ILS (NOTE 3)	MKR SENS 1=LOW	GLS /NO 1=NO	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	◀-----LSB]				PAD	MARKER BEACONS		
	◀-----]					INN	MDL	OUT
	◀-----GS DEV-----]					1=ON	1=ON	1=ON
	(NOTE 2)							
(3)	[MSB-----2'S COMPLEMENT,12-BIT-----◀-----							
	[-0.80-----DDM,%/100-----◀-----							
	[FLY DWN-----GLIDESLOPE DEVIATION-----◀-----							
(4)	◀-----LSB]				PAD	PAD	PAD	PAD
	◀-----]							
	◀-----LOC DEV-----]							
(5)	[MSB-----2'S COMPLEMENT,12-BIT-----◀-----							
	[-0.40-----DDM,%/100-----◀-----							
	[FLY LEFT-----LOCALIZER DEVIATION-----◀-----							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

NOTE 2: MARKER BEACON BITS (2,1,0) BYTE (2)

THESE 1-BIT MESSAGES ARE TRANSMITTED UPON MARKER RECEPTION
INDEPENDENTLY OF THE STATE OF THE FLAG BITS (7,6) BYTE (1)

NOTE 3: DELAYED ILS IS NOW AVAILABLE IN CURRENT PRODUCTION RADIOS.

TITLE: DME FREQUENCY

ADDRESS 24

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0 0 1 0] [-----HEX 2-----] [-----ADDRESS-----]				[0 1 0 0] [-----HEX 4-----] [-----BYTE-----]			
(1)	FREQ VALID 1=VAL	AUTO/ MANUAL TUNE (NOTE 2)	PAD	MLS/ VORLOC IDENT 1=MLS	DME HOLD 1=ON	TEST 1=ON	SI CHANNEL IDENT (NOTE 1)	SI
(2)	[8 4 2 1] [-----BCD-----] [-----0.1 MHZ-----]				[8 4 2 1] [-----BCD-----] [-----0.01 MHZ-----]			
(3)	MLS CHAN IDENT (NOTE 3)	[4 2 1] [-----BCD-----] [---10.0 MHZ-----]			[8 4 2 1] [-----BCD-----] [-----1.0 MHZ-----]			
(4)	[PAD PAD PAD PAD] [-----] [-----]				[PAD PAD PAD PAD] [-----] [-----]			
(5)	RMT TUNE 1=INHIB	AUTO TUNE INHIB 1=INHIB	[PAD PAD] [-----] [-----]		[PAD PAD PAD PAD] [-----] [-----]			

NOTE 1: CHANNEL IDENT BITS (1,0) BYTE (1)

1	0	
0	0	ON-SIDE PRESET (OR REMOTE)
0	1	#1 ACTIVE
1	0	#2 ACTIVE
1	1	REMOTE ONLY

COMMENT: ON-SIDE PRESET IS PRE-EMPTED BY REMOTE TUNING COMMAND ON ADDRESS LABEL 34 (#1 OR #2 SIDE PRE-EMPT CHANNEL IDENT)

NOTE 2: AUTO/MANUAL TUNE BIT(6) BYTE(1)

0 = AUTO TUNE DISABLE (MANUAL TUNE)
1 = AUTO (DATA BASE) TUNE ABLE

NOTE 3: MLS CHANNEL IDENT BIT(7) BYTE(3)

0 = X/Y TACAN CHANNEL (EVEN NUMBERED MLS)
1 = W/Z TACAN CHANNEL (ODD NUMBERED MLS)

TITLE: DME FREQUENCY & DISTANCE

ADDRESS 25

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	0]	[0	1	0	1]
	[-----HEX 2-----]				[-----HEX 5-----]			
	[-----ADDRESS BYTE-----]							
(1)	FREQ VALID 1=VAL	DIST VALID 1=VAL	SRCH/ TRCK *	SRCH/ TRCK *	DME HOLD 1=ON	TEST 1=ON	SI CHANNEL IDENT	SI (NOTE 1)
	(NOTE 2)							
(2)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.1 MHZ-----]				[-----0.01 MHZ-----]			
(3)	MLS CHAN IDENT (NOTE 3)	[4	2	1]	[8	4	2	1]
		[---BCD-----]				[-----BCD-----]		
		[--10.0 MHZ-----]				[-----1.0 MHZ-----]		
(4)	[MSB-----BINARY, 16-BIT-----LSB]							
	[-----NAUTICAL MILES-----]							
	[-----DME DISTANCE-----]							
(5)	[MSB-----BINARY 16-BIT-----LSB]							
	[256-----NAUTICAL MILES-----]							
	[-----DME DISTANCE-----]							

NOTE 1: CHANNEL IDENT: (BITS 1,0)

$\frac{1}{0} \frac{0}{0}$ ON-SIDE PRESET (OR REMOTE)
 0 1 #1 ACTIVE
 1 0 #2 ACTIVE
 1 1 REMOTE ONLY
 COMMENT: ON-SIDE PRESET IS PRE-EMPTED BY REMOTE TUNING COMMAND ON
 ADDRESS LABEL 34 (#1 OR #2 SIDE PRE-EMPT CHANNEL IDENT)

NOTE 2: *SRCH/TRCK: (BITS 5,4)

$\frac{5}{0} \frac{4}{0}$ NO SQUITTER
 0 1 SEARCH
 1 0 PRE-TRACK
 1 1 TRACK

NOTE 3: MLS CHANNEL IDENT BIT(7) BYTE(3)

0 = X/Y TACAN CHANNEL (EVEN NUMBERED MLS)
 1 = W/Z TACAN CHANNEL (ODD NUMBERED MLS)

ADDRESS 26 TITLE: DME TIME-TO-STATION & VELOCITY

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	0]	[0	1	1	0]
	[-----HEX 2-----]				[-----HEX 6-----]			
	[-----ADDRESS BYTE-----]							
(1)	TTS & VEL 1=VAL	AUTO/ MANUAL TUNE (NOTE 4)	SRCH/ TRCK * (NOTE 2)	SRCH/ TRCK * (NOTE 2)	MLS/ VORLOC IDENT 1=MLS	TEST 1=ON	SI CHANNEL IDENT (NOTE 1)	SI
(2)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----1.0 MIN-----]				[-----0.10 MIN-----]			
(3)	[PAD	PAD	PAD	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----100 MIN-----]				[-----10 MIN-----]			
(4)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----10 KNOTS-----]				[-----1.0 KNOTS-----]			
(5)	RMT TUNE 1=INHIB	AUTO TUNE INHIB 1=INHIB	PAD	PAD]	[8	4	2	1]
					[-----BCD-----]			
					[-----100 KNOTS-----]			

NOTE 1: CHANNEL IDENT: (BITS 1,0)

1 0
 0 0 ON-SIDE PRESET (OR REMOTE)
 0 1 #1 ACTIVE
 1 0 #2 ACTIVE
 1 1 REMOTE ONLY

COMMENT: ON-SIDE PRESET IS PRE-EMPTED BY REMOTE TUNING COMMAND ON ADDRESS LABEL 34 (#1 OR #2 SIDE PRE-EMPT CHANNEL IDENT)

NOTE 2: *SRCH/TRCK: (BITS 5,4)

5 4
 0 0 NO SQUITTER
 0 1 SEARCH
 1 0 PRE-TRACK
 1 1 TRACK

NOTE 3: TIME-TO-STATION & VELOCITY BOTH ARE COMPUTED AND PRESENT ON OUTPUT BUS WITHOUT REGARD TO SIGN.

NOTE 4: AUTO/MANUAL TUNE BIT(6) BYTE1)

0 = AUTO TUNE DISABLE (MANUAL TUNE)
 1 = AUTO (DATA BASE) TUNE ENABLE

TITLE: DME IDENT

ADDRESS 27

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	0]	[0	1	1	1]
	[-----HEX 2-----]				[-----HEX 7-----]			
	[-----ADDRESS BYTE-----]							
(1)	IDENT VALID 1=VAL	ANALG & 568 DATA (NOTE 3)	SRCH/ TRCK * (NOTE 2)	SRCH/ TRCK *	PAD	TEST 1=ON	SI CHANNEL IDENT (NOTE 1)	SI
(2)	0=VAL CHAR VALID	[1	0	0	1	0	0	1]
		[7-BIT ASCII-----EXAMPLE=I -----]						
		[-----FIRST CHARACTER--(NOTE 4)-----]						
(3)	0=VAL CHAR VALID	[1	0	0	1	1	0	0]
		[7-BIT ASCII-----EXAMPLE=L -----]						
		[-----SECOND CHARACTER-----]						
(4)	0=VAL CHAR VALID	[1	0	0	0	0	0	1]
		[7-BIT ASCII-----EXAMPLE=A-----]						
		[-----THIRD CHARACTER-----]						
(5)	0=VAL CHAR VALID	[1	0	1	1	0	0	0]
		[7-BIT ASCII-----EXAMPLE=X-----]						
		[-----FOURTH CHARACTER-----]						

NOTE 1: CHANNEL IDENT: (BITS 1,0)

1 0
 0 0 ON-SIDE PRESET (OR REMOTE)
 0 1 #1 ACTIVE
 1 0 #2 ACTIVE
 1 1 REMOTE ONLY
 COMMENT: ON-SIDE PRESET IS PRE-EMPTED BY REMOTE TUNING COMMAND ON
 ADDRESS LABEL 34 (#1 OR #2 SIDE PRE-EMPT CHANNEL IDENT)

NOTE 2: *SRCH/TRCK: (BITS 5,4)

5 4
 0 0 NO SQUITTER
 0 1 SEARCH
 1 0 PRE-TRACK
 1 1 TRACK

NOTE 3: ANALOG AND 568 DATA (BIT 6 BYTE 1)

A "1" IN THIS BIT IDENTIFIES THE CHANNEL WHICH IS THE SOURCE OF DATA FOR THE ANALOG AND 568 6-WIRE OUTPUTS.

NOTE 4: A 3-LETTER IDENT OCCUPIES THE 2ND, 3RD, & 4TH CHARACTER BYTES, WITH THE 1ST CHARACTER BYTE (2), BIT (7) SET INVALID.

TITLE: STEER COMMANDS/ALT REFERENCE

ADDRESS 2A

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	0]	[1	0	1	0]
	[-----HEX 2-----]				[-----HEX A-----]			
	[-----ADDRESS BYTE-----]							
(1)	ALT REF 1=VAL	PITCH STEER 1=VAL	ROLL STEER 1=VAL	PAD	PAD	RSVD TEST	RESERVED SOURCE IDENT	
(2)	◀-----2'S COMPLEMENT, 16-BIT-----▶ ◀-----FEET-----▶ ◀-----ALTITUDE REFERENCE-----▶							
(3)	[MSB----- (BIASED +24576 FEET) -----▶ [-32768----- FEET -----▶ [----- ALTITUDE REFERENCE -----▶							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----▶ ◀-----DEGREES-----▶ ◀-----PITCH STEER-----▶							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----▶ [-180-----DEGREES-----▶ [FLY-DOWN-----PITCH-STEER-----▶							
(6)	◀-----2'S COMPLEMENT, 16-BIT-----▶ ◀-----DEGREES-----▶ ◀-----ROLL STEER-----▶							
(7)	[MSB-----2'S COMPLEMENT, 16-BIT-----▶ [-180-----DEGREES-----▶ [FLY LEFT-----ROLL STEER-----▶							

TITLE: ADF DATA

ADDRESS 2B

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	0]	[1	0	1	1]
	[-----HEX 2-----]				[-----HEX B-----]			
	[-----ADDRESS BYTE-----]							
(1)	ADF BRNG 1=VAL	FREQ VALID 1=VAL	BFO 1=ON	ADF/ ANT 1=ADF	FREQ LIMIT 1=ON	TEST 1=ON	SI SOURCE IDENT	SI (NOTE 1)
					(NOTE 2)			
(2)	<-----2'S COMPLEMENT, 12-BIT-----LSB]							
	<-----DEGREES-----]							
	<-----ADF BEARING-----]							
(3)	[8	4	2	1]	[MSB-----	<-----		
	[-----BCD-----]				[-180 -----<-----			
	[-----O.1 KHZ-----]				[-----ADF BEARING-----<-----			
(4)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----10 KHZ-----]				[-----1.0 KHZ-----]			
(5)	[PAD	PAD]	[2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----1000 KHZ-----]				[-----100 KHZ-----]			

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

NOTE 2: FREQUENCY LIMIT BIT (3) BYTE (1)

1=1750 KHZ MAX

TITLE: ADF FREQUENCY

ADDRESS 2C

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	0]	[1	1	0	0]
	[-----HEX 2-----]				[-----HEX C-----]			
	[-----ADDRESS BYTE-----]							
(1)	FREQ VALID 1=VAL	PAD	BFO 1=ON	ADF/ ANT 1=ADF	PAD	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	[PAD	PAD	PAD	PAD	PAD	PAD	PAD	PAD]
	[-----]							
	[-----]							
(3)	[8	4	2	1]	[PAD	PAD	PAD	PAD]
	[-----BCD-----]							
	[-----0.1 KHZ-----]							
(4)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----10 KHZ-----]				[-----1.0 KHZ-----]			
(5)	[PAD	PAD]	[2	1]	[8	4	2	1]
			[----BCD----		[-----BCD-----]			
			[-1000 KHZ--]		[-----100 KHZ-----]			

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

TITLE: REMOTE NAV TUNE

ADDRESS 30

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	1]	[0	0	0	0]
	[-----HEX 3-----]				[-----HEX 0-----]			
	[-----ADDRESS BYTE-----]							
(1)	REM TUNE 1=VAL	AUTO/ MANUAL TUNE (NOTE 3)	ACT /PRE 1=XFR (NOTE 2)	REM MEM 1=RCL	MLS/ VORLOC IDENT 1=MLS	TEST 1=ON (NOTE 2)	SI SOURCE IDENT (NOTE 1)	SI
(2)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.1 MHZ-----]				[-----0.01 MHZ-----]			
(3)	MLS CHAN IDENT (NOTE 4)	[4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----10.0 MHZ-----]				[-----1.0 MHZ-----]			
(4)	RMT TUNE 1=INHIB	AUTO TUNE INHIB 1=INHIB	[PAD PAD]	[PAD PAD]	[PAD PAD]	[PAD PAD]	[PAD PAD]	[PAD PAD]
	[-----]				[-----]			
(5)	[-----2'S COMPLEMENT-----]							
	[-----MODULO 256-----]							
	[-----CHECKSUM-----]							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	SET #1 PRESET
0	1	SET #1 ACTIVE
1	0	SET #2 ACTIVE
1	1	SET #2 PRESET

NOTE 2: CONTROL BITS (5,4,2) BYTE (1)

THESE ARE MOMENTARY ACTION CONTROL BITS, AND REQUIRE BOTH SET AND RESET STATES TO BE TRANSMITTED EACH TIME. SEE COMMENTS UNDER BUS OPERATIONS PARAGRAPH ABOVE.

NOTE 3: AUTO/MANUAL TUNE BIT(6) BYTE(1)

0 = AUTO TUNE DISABLE (MANUAL TUNE)
1 = AUTO (DATA BASE) TUNE ABLE

NOTE 4: MLS CHANNEL IDENT BIT(7) BYTE(3)

0 = X/Y TACAN CHANNEL (EVEN NUMBERED MLS)
1 = W/Z TACAN CHANNEL (ODD NUMBERED MLS)

ADDRESS 31 TITLE: REMOTE COMM TUNE

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	1]	[0	0	0	1]
	[-----HEX 3-----]				[-----HEX 1-----]			
	[-----ADDRESS BYTE-----]							
(1)	REM TUNE 1=VAL	PAD	ACT /PRE 1=XFR	REM MEM 1=RCL (NOTE 2)	SI (NOTE 1)	TEST 1=ON (NOTE 2)	SI SOURCE IDENT (NOTE 1)	SI
(2)	[8	4	2	1]	[PAD	PAD	PAD	PAD]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.001 MHZ-----]				[-----BCD-----]			
(3)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.1 MHZ-----]				[-----0.01 MHZ-----]			
(4)	[PAD	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----10.0 MHZ-----]				[-----1.0 MHZ-----]			
(5)	[-----2'S COMPLEMENT-----]							
	[-----MODULO 256-----]							
	[-----CHECKSUM-----]							

NOTE 1: SOURCE IDENT BITS (3,1,0) BYTE (1)

3 1 0		3 1 0
0 0 0	NOT USED	1 0 0 NOT USED
0 0 1	#1 ACTIVE	1 0 1 #1 PRESET
0 1 0	#2 ACTIVE	1 1 0 #2 PRESET
0 1 1	#3 ACTIVE	1 1 1 #3 PRESET

NOTE 2: CONTROL BITS (5,4,2) BYTE (1)

THESE ARE MOMENTARY ACTION CONTROL BITS, AND REQUIRE BOTH SET AND RESET STATES TO BE TRANSMITTED EACH TIME. SEE COMMENTS UNDER BUS OPERATIONS PARAGRAPH ABOVE.

TITLE: REMOTE ADF TUNE

ADDRESS 32

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	1]	[0	0	1	0]
	[-----HEX 3-----]				[-----HEX 2-----]			
	[-----ADDRESS BYTE-----]							
(1)	REM TUNE 1=VAL	PAD	ACT /PRE 1=XFR	REM MEM 1=RCL (NOTE 2)	PAD	TEST 1=ON (NOTE 2)	SI SOURCE IDENT (NOTE 1)	SI
(2)	[8	4	2	1]	[PAD	PAD	PAD	PAD]
	[-----BCD-----]							
	[-----O.1 KHZ-----]							
(3)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----10 KHZ-----]				[-----1.0 KHZ-----]			
(4)	[PAD	PAD]	[2	1]	[8	4	2	1]
			[---BCD---		[-----BCD-----]			
			[1000 KHZ---		[-----100 KHZ-----]			
(5)	[-----2'S COMPLEMENT-----]							
	[-----MODULO 256-----]							
	[-----CHECKSUM-----]							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	SET #1 PRESET
0	1	SET #1 ACTIVE
1	0	SET #2 ACTIVE
1	1	SET #2 PRESET

NOTE 2: CONTROL BITS (5,4,2) BYTE (1)

THESE ARE MOMENTARY ACTION CONTROL BITS, AND REQUIRE BOTH SET AND RESET STATES TO BE TRANSMITTED EACH TIME. SEE COMMENTS UNDER BUS OPERATIONS PARAGRAPH ABOVE.

TITLE: REMOTE ATC CODE

ADDRESS 33

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	1]	[0	0	1	1]
	[-----HEX 3-----]				[-----HEX 3-----]			
	[-----ADDRESS				BYTE-----]			
(1)	CODE	PAD	PAD	PAD	PAD	TEST	SI	SI
	VALID						SOURCE	
	1=VAL					1=ON	IDENT	
							(NOTE 1)	
(2)	[PAD	4	2	1]	[PAD	4	2	1]
	[-----ATC CODE C-----]				[-----ATC CODE D-----]			
	[-----BCD, (0-7 ONLY)-----]							
(3)	[PAD	4	2	1]	[PAD	4	2	1]
	[-----ATC CODE A-----]				[-----ATC CODE B-----]			
	[-----BCD, (0-7 ONLY)-----]							
(4)	[PAD	PAD	PAD	PAD]	[PAD	PAD	PAD	PAD]
	[-----]				[-----]			
	[-----]				[-----]			
(5)	[-----2'S COMPLEMENT-----]							
	[-----MODULO 256-----]							
	[-----CHECKSUM-----]							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

<u>1</u>	<u>0</u>	
0	0	NOT USED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	NOT USED

TITLE: REMOTE DME TUNE

ADDRESS 34

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	1]	[0	1	0	0]
	[-----HEX 3-----]				[-----HEX 4-----]			
	[-----ADDRESS BYTE-----]							
(1)	REM TUNE 1=VAL	AUTO/ MANUAL TUNE (NOTE 2)	PAD	MLS/ VORLOC IDENT 1=MLS	PAD	RSVD	SI CHANNEL IDENT (NOTE 1)	SI
(2)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.1 MHZ-----]				[-----0.01 MHZ-----]			
(3)	MLS CHAN IDENT (NOTE 3)	[4	2	1]	[8	4	2	1]
	[----BCD-----]				[----BCD-----]			
	[--10.0 MHZ-----]				[-----1.0 MHZ-----]			
(4)	RMT TUNE 1=INHIB	AUTO TUNE INHIB 1=INHIB	[PAD	PAD]	[PAD	PAD	PAD	PAD]
	[-----]				[-----]			
(5)	[-----2'S COMPLEMENT-----]							
	[-----MODULO 256-----]							
	[-----CHECKSUM-----]							

NOTE 1: CHANNEL IDENT BITS (1,0) BYTE (1)

1	0	
0	0	#1 SIDE PRE-EMPT
0	1	#1 REMOTE
1	0	#2 REMOTE
1	1	#2 SIDE PRE-EMPT

(NOTE 1 CONTINUES ON NEXT PAGE)

NOTE 1: (CONTINUED)

PURPOSE:

THIS ADDRESS IS TO BE USED ONLY TO TUNE A DME CHANNEL TO A FREQUENCY INDEPENDENT OF THE ACTIVE, ASSOCIATED NAV CHANNEL. AVAILABLE CHANNELS ARE AS DESCRIBED BELOW:

DUAL DME INSTALLATIONS:

ONE DME CHANNEL IN EACH DME RECEIVER IS DEDICATED TO ITS ASSOCIATED ACTIVE NAV CHANNEL, AND IN ADDITION, A SECOND CHANNEL IN EACH DME IS ASSIGNED TO ITS ASSOCIATED PRESET NAV CHANNEL, LEAVING TWO DME CHANNELS AVAILABLE FOR FULL-TIME, REMOTE USE. THESE TWO CHANNELS ARE TUNED USING ADDRESS 34 AND SOURCE IDENT BITS (0,1) FOR #1 DME AND (1,0) FOR #2 DME. IN ADDITION, TUNING VIA THIS ADDRESS IS ALLOWED TO "PREEMPT" THE TWO CHANNELS NORMALLY TUNED TO THE PRESET NAV CHANNELS, BY USING IDENT BITS (0,0) FOR #1 DME AND (1,1) FOR #2 DME. NOTE THAT THE REMOTE BUS THEN REMAINS IN CONTROL OF THAT CHANNEL UNTIL A NEW BURST IS SENT WITH THE 34 LABEL, THE SAME IDENT BITS, (0,0 OR 1,1), & THE REM TUNE VALID, BIT (7) = 0, OR UNTIL THE SYSTEM IS POWERED DOWN. THIS OPERATION IS NEARLY TRANSPARENT TO THE PILOT.

SINGLE DME ON THE #1 SIDE:

TWO DME CHANNELS ARE DEDICATED TO THE TWO ACTIVE NAV CHANNELS, AND ONLY CHANNEL (0,0) IS AVAILABLE FOR REMOTE TUNING. IT OPERATES BY PRE-EMPTING THE #1 PRESET CHANNEL AS DESCRIBED ABOVE. IF THE OTHER REMOTE SOURCE IDENT CODES (1,0), (0,1), (1,1) ARE SENT, THEY WILL BE IGNORED.

SINGLE DME ON THE #2 SIDE:

NOT A RECOMMENDED INSTALLATION.

NOTE 2: AUTO/MANUAL TUNE BIT(6) BYTE(1)

0 = AUTO TUNE DISABLE (MANUAL TUNE)
1 = AUTO (DATA BASE) TUNE ABLE

NOTE 3: MLS CHANNEL IDENT BIT(7) BYTE(3)

0 = X/Y TACAN CHANNEL (EVEN NUMBERED MLS)
1 = W/Z TACAN CHANNEL (ODD NUMBERED MLS)

ADDRESS 35 TITLE: REMOTE COMM TUNE (8.33 KHZ CHANNELS)

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	1]	[0	1	0	1]
	[-----HEX 3-----]				[-----HEX 5-----]			
	[-----ADDRESS BYTE-----]							
(1)	REM TUNE 1=VAL	PAD	ACT /PRE 1=XFR (NOTE 2)	REM MEM 1=RCL	SI (NOTE 1)	TEST 1=ON (NOTE 2)	SI SOURCE IDENT (NOTE 1)	SI
(2)	[8	4	2	1]	[PAD	PAD	PAD	PAD]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.001 MHZ-----]				[-----BCD-----]			
(3)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----0.1 MHZ-----]				[-----0.01 MHZ-----]			
(4)	[PAD	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----10.0 MHZ-----]				[-----1.0 MHZ-----]			
(5)	[-----2'S COMPLEMENT-----]							
	[-----MODULO 256-----]							
	[-----CHECKSUM-----]							

NOTE 1: SOURCE IDENT BITS (3,1,0) BYTE (1)

<u>3</u> <u>1</u> <u>0</u>		<u>3</u> <u>1</u> <u>0</u>
0 0 0	NOT USED	1 0 0 NOT USED
0 0 1	#1 ACTIVE	1 0 1 #1 PRESET
0 1 0	#2 ACTIVE	1 1 0 #2 PRESET
0 1 1	#3 ACTIVE	1 1 1 #3 PRESET

NOTE 2: CONTROL BITS (5,4,2) BYTE (1)

THESE ARE MOMENTARY ACTION CONTROL BITS, AND REQUIRE BOTH SET AND RESET STATES TO BE TRANSMITTED EACH TIME. SEE COMMENTS UNDER BUS OPERATIONS PARAGRAPH ABOVE.

TITLE: REMOTE MLS CHANNEL AND GLIDE PATH TUNE

ADDRESS 37

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	1]	[0	1	1	1]
	[-----HEX 3-----]				[-----HEX 7-----]			
	[-----ADDRESS BYTE-----]							
(1)	REM TUNE 1=VAL	MAN/ AUTO VALID 1=VAL	PAD	MAN/ AUTO MODE 1=AUTO	AUTO/ KBD TUNE 1=AUTO	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----CHANNEL MID-----]				[-----CHANNEL LOW-----]			
(3)	[8	4	2	1]	CH VALID 1=VAL	[4	2	1]
	[-----BCD-----]					[-----BCD-----]		
	[----GLIDE PATH LOW-----]					[---CHANNEL HIGH---		
(4)	GP VALID 1=VAL	[PAD	PAD	1]	[8	4	2	1]
	[---BCD-----]				[-----BCD-----]			
	[-GLIDE PATH HIGH-]				[-----GLIDE PATH MID-----]			
(5)	[-----2'S COMPLEMENT-----]							
	[-----MODULO 256-----]							
	[-----CHECKSUM-----]							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1 0	
0 0	NOT USED OR NOT DEFINED
0 1	#1 ACTIVE
1 0	#2 ACTIVE
1 1	RESERVED

TITLE: REMOTE MLS AZIMUTH AND BACK AZIMUTH TUNE

ADDRESS 38

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	0	1	1]	[1	0	0	0]
	[-----HEX 3-----]				[-----HEX 8-----]			
	[-----ADDRESS BYTE-----]							
(1)	REM TUNE 1=VAL	AZ/ BAZ MODE VALID 1=VAL	PAD	AZ/ BAZ MODE 1=BAZ	PAD	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	[8	4	2	1]	[8	4	2	1]
	[-----BCD-----]				[-----BCD-----]			
	[-----AZIMUTH MID-----]				[-----AZIMUTH LOW-----]			
(3)	[8	4	2	1]	AZ VALID 1=VAL	[PAD	2	1]
	[-----BCD-----]					[-----BCD-----]		
	[---BACK AZIMUTH LOW---					[---AZIMUTH HIGH---		
(4)	BAZ VALID 1=VAL	[PAD	2	1]	[8	4	2	1]
		[---BCD-----]			[-----BCD-----]			
		[---BACK AZ HIGH---			[-----BACK AZIMUTH MID---			
(5)	[-----2'S COMPLEMENT-----]							
	[-----MODULO 256-----]							
	[-----CHECKSUM-----]							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED OR NOT DEFINED
0	1	#1 ACTIVE
1	0	#2 ACTIVE
1	1	RESERVED

TITLE: RATES, (BODY AXIS)

ADDRESS 40

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	1	0	0]	[0	0	0	0]
	[-----HEX 4-----]				[-----HEX 0-----]			
	[-----ADDRESS BYTE-----]							
(1)	DATA VALID 1=VAL	PAD	DG MODE 1=ON	STRUT SW 1=GND	SI	TEST 1=ON (NOTE 1)	SI SOURCE IDENT (NOTE 1)	SI
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES/SECOND-----]							
	◀-----PITCH RATE-----]							
(3)	[MSB-----BODY-AXIS-----◀							
	[-256-----DEGREES/SECOND-----◀							
	[NOSE DOWN-----PITCH RATE-----◀							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES/SECOND-----]							
	◀-----ROLL RATE-----]							
(5)	[MSB-----BODY-AXIS-----◀							
	[-256-----DEGREES/SECOND-----◀							
	[NOSE LEFT-----ROLL RATE-----◀							
(6)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES/SECOND-----]							
	◀-----YAW RATE-----]							
(7)	[MSB-----BODY AXIS-----◀							
	[-256-----DEGREES/SECOND-----◀							
	[NOSE LEFT-----YAW RATE -----◀							

NOTE 1: SOURCE IDENT BITS (3,1,0) BYTE (1)

3 1 0		3 1 0
0 0 0	NOT USED	1 0 0 NOT USED
0 0 1	#1 AHC	1 0 1 #1 IRU
0 1 0	#2 AHC	1 1 0 #2 IRU
0 1 1	#3 AHC	1 1 1 #3 IRU

ADDRESS 41 TITLE: ACCELERATIONS, (BODY AXIS)

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0 1 0 0]				[0 0 0 1]			
	[-----HEX 4-----]				[-----HEX 1-----]			
	[-----ADDRESS BYTE-----]							
(1)	DATA VALID 1=VAL	PAD	DG MODE 1=ON	STRUT SW 1=GND	SI	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----" G'S "-----]							
	◀-----LONGITUDINAL ACCELERATION-----]							
(3)	[MSB-----BODY AXIS-----◀							
	[-10-----" G'S "-----◀							
	[AFT-----LONGITUDINAL ACCELERATION-----◀							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----" G'S "-----]							
	◀-----LATERAL ACCELERATION-----]							
(5)	[MSB-----BODY AXIS-----◀							
	[-10-----" G'S "-----◀							
	[LEFT-----LATERAL ACCELERATION-----◀							
(6)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----" G'S "-----]							
	◀-----NORMAL ACCELERATION-(NOTE 2)-----]							
(7)	[MSB-----BODY AXIS-----◀							
	[-10-----" G'S "-----◀							
	[NOSE DOWN-----NORMAL ACCELERATION-(NOTE 2)-----◀							

NOTE 1: SOURCE IDENT BITS (3,1,0) BYTE (1)

3 1 0		3 1 0
0 0 0	NOT USED	1 0 0 NOT USED
0 0 1	#1 AHC	1 0 1 #1 IRU
0 1 0	#2 AHC	1 1 0 #2 IRU
0 1 1	#3 AHC	1 1 1 #3 IRU

NOTE 2: NORMAL ACCELERATION DATA VALUE IS 1 G FOR NO EXTERNAL FORCES, OTHER THAN EARTH'S GRAVITY, ON AHC/IRU.

ADDRESS 42 TITLE: ACCELERATIONS

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	1	0	0]	[0	0	1	0]
	[-----HEX 4-----]				[-----HEX 2-----]			
	[-----ADDRESS BYTE-----]							
(1)	DATA VALID 1=VAL	PAD	DG MODE 1=ON	STRUT SW 1=GND	SI (NOTE 1)	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	<-----2'S COMPLEMENT, 16-BIT-----LSB]							
	<-----" G'S "-----]							
	<-----ALONG HDG ACCELERATION-----]							
(3)	[MSB-----							
	[-10-----" G'S "-----							
	[AFT-----ALONG HDG ACCELERATION-----							
(4)	<-----2'S COMPLEMENT, 16-BIT-----LSB]							
	<-----" G'S "-----]							
	<-----CROSS HDG ACCELERATION-----]							
(5)	[MSB-----							
	[-10-----" G'S "-----							
	[LEFT-----CROSS HDG ACCELERATION-----							
(6)	<-----2'S COMPLEMENT, 16-BIT-----LSB]							
	<-----" G'S "-----]							
	<-----VERTICAL ACCELERATION-----]							
(7)	[MSB-----							
	[-10-----" G'S "-----							
	[DOWN-----VERTICAL ACCELERATION-----							

NOTE 1: SOURCE IDENT BITS (3,1,0) BYTE (1)

3 1 0		3 1 0
0 0 0	NOT USED	1 0 0 NOT USED
0 0 1	#1 AHC	1 0 1 #1 IRU
0 1 0	#2 AHC	1 1 0 #2 IRU
0 1 1	#3 AHC	1 1 1 #3 IRU

TITLE: ATTITUDE/HEADING (EULER)

ADDRESS 43

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	1	0	0]	[0	0	1	1]
	[-----HEX 4-----]				[-----HEX 3-----]			
	[-----ADDRESS BYTE-----]							
(1)	ATT VALID 1=VAL	HDG VALID 1=VAL	RSVD REV MODE2	REV MODEL 1=ON	SI (NOTE 1)	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES-----]							
	◀-----PITCH ATTITUDE-----]							
(3)	[MSB-----◀							
	[-180-----DEGREES-----◀							
	[NOSE DOWN-----PITCH ATTITUDE-----◀							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES-----]							
	◀-----ROLL ATTITUDE-----]							
(5)	[MSB-----◀							
	[-180-----DEGREES-----◀							
	[LEFT-----ROLL ATTITUDE-----◀							
(6)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES-----]							
	◀-----MAGNETIC HEADING-----]							
(7)	[MSB-----◀							
	[-180-----DEGREES-----◀							
	[SOUTH-----MAGNETIC HEADING-----◀							

NOTE 1: SOURCE IDENT BITS (3,1,0) BYTE (1)

$\frac{3}{0}$	$\frac{1}{0}$	$\frac{0}{0}$	NOT USED	$\frac{3}{1}$	$\frac{1}{0}$	$\frac{0}{0}$	NOT USED
0	0	1	#1 AHC	1	0	1	#1 IRU
0	1	0	#2 AHC	1	1	0	#2 IRU
0	1	1	#3 AHC	1	1	1	#3 IRU

ADDRESS 46		TITLE: HEADING, MAGNETIC							
BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0	

(0)	[0 1 0 0]				[0 1 1 0]				
	[-----HEX 4-----]				[-----HEX 6-----]				
	[-----ADDRESS BYTE-----]								
(1)	HDG VAL 1=VAL	PAD	PAD	PAD	SLAVE MODE 1=ON	RSVD TEST	RESERVED SOURCE IDENT		
(2)	◀-----2'S COMPLEMENT, 14-BIT-----LSB]						PAD	PAD	
	◀-----DEGREES-----]								
	◀-----MAGNETIC HEADING-----]								
(3)	[MSB-----LSB]								◀-
	[-180-----DEGREES-----]								◀-
	[SOUTH-----MAGNETIC HEADING-----]								◀-
(4)	[MSB-----2'S COMPLEMENT, 8-BIT-----LSB]								
	[-MAX-----(-MAX, 0, +MAX ONLY)-----]								
	[DECR HDG-----SLAVE RATE-----]								
(5)	[MSB-----LSB]								
	[TBD-----TBD-----]								
	[-----SPARE-----]								

TITLE: AHRS CONTROL

ADDRESS 47

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	1	0	0]	[0	1	1	1]
	[-----HEX 4-----]				[-----HEX 7-----]			
	[-----ADDRESS BYTE-----]							
(1)	ATT VALID 1=VAL	HDG VALID 1=VAL	RSVD REV MODE2	REV MODE1 1=ON	SI (NOTE 1)	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]◀-----DEGREES-----]◀-----MAGNETIC HEADING-----]							
(3)	[MSB-----◀-----DEGREES-----◀-----MAGNETIC HEADING-----◀-----							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]◀-----DEGREES-----]◀-----LATITUDE-----]							
(5)	[MSB-----◀-----DEGREES-----◀-----LATITUDE-----◀-----							
(6)	[MSB-----2'S COMPLEMENT, 8-BIT-----LSB]◀-----DEGREES DIFFERENCE-----]◀-----COMPASS CARD SLAVING-----]							
(7)	[8	4	2	1]	[8	4	2	1]
	[-----HEX-----]				[-----HEX-----]			
	[---2-DIGIT STIMULUS MODE CYCLE IDENT (NOTE 2)-----]							

NOTE 1: SOURCE IDENT BITS (3,1,0) BYTE (1)

3 1 0		3 1 0		3 1 0	
0 0 1	#1 AHC	1 0 1	#1 IRU	0 0 0	NOT USED
0 1 0	#2 AHC	1 1 0	#2 IRU	1 1 1	NOT USED
0 1 1	#3 AHC	1 1 1	#3 IRU		

NOTE 2: STIM MODE IDENT BYTE (7), HEX DIGITS

00 STIM OFF	04 NOT USED	08 NOT USED
01 ROLL ONLY	05 NOT USED	09 NOT USED
02 PITCH ONLY	06 NOT USED	0A ALL DATA
03 HDG ONLY	07 NOT USED	CYCLING

TITLE: CRT ASCII MESSAGES

ADDRESS 70

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	1	1	1]	[0	0	0	0]
	[-----HEX 7-----]				[-----HEX 0-----]			
	[-----ADDRESS BYTE-----]							
(1)	[---BINARY,3-BIT---]				[-----BINARY,5-BIT-----]			
	[-----LINE-----]				[-----COLUMN-----]			
	[---IDENTIFIER---]				[-----IDENTIFIER-----]			
(2)	[-----8-BIT ASCII-----]							
	[----- (LEFT MOST) -----]							
	[-----CHARACTER #1-----]							
(3)	[-----8-BIT ASCII-----]							
	[-----CHARACTER #2-----]							
(4)	[-----8-BIT ASCII-----]							
	[-----CHARACTER #3-----]							
(5)	[-----8-BIT ASCII-----]							
	[-----CHARACTER #4-----]							
(6)	[-----8-BIT ASCII-----]							
	[-----CHARACTER #5-----]							
(7)	[-----8-BIT ASCII-----]							
	[----- (RIGHT MOST) -----]							
	[-----CHARACTER #6-----]							

TITLE: VS/IAS/MACH REFERENCES

ADDRESS 72

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	1	1	1]	[0	0	1	0]
	[-----HEX 7-----]				[-----HEX 2-----]			
	[-----ADDRESS BYTE-----]							
(1)	VS REF	IAS REF	MACH REF	VS BUG	IAS BUG	REF BUG CONT	SI SOURCE	SI IDENT
	[1=SET DISPLAY ON]			[1 = SLAVE]		1=DSBL (NOTE 2)	(NOTE 1)	
(2)	<-----2'S COMPLEMENT,16-BIT----->LSB]							
	<-----FEET/MINUTE----->]							
	<-----VERTICAL SPEED REFERENCE----->]							
(3)	[MSB----->]							
	[-20480----->FEET/MINUTE----->]							
	[DOWN----->VERTICAL SPEED REFERENCE----->]							
(4)	<-----2'S COMPLEMENT, 16-BIT----->LSB]							
	<-----KNOTS----->]							
	<-----IAS REFERENCE----->]							
(5)	[MSB----->2'S COMPLEMENT,16-BIT----->]							
	[-512 ----->KNOTS----->]							
	[----->IAS REFERENCE----->]							
(6)	<-----2'S COMPLEMENT, 16-BIT----->LSB]							
	<-----NUMBER----->]							
	<-----MACH REFERENCE----->]							
(7)	[MSB----->2'S COMPLEMENT,16-BIT----->]							
	[-1.024----->NUMBER----->]							
	[----->MACH REFERENCE----->]							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED OR NOT DEFINED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

NOTE 2: BIT ONLY HAS MEANING WHEN BIT 3, OF BYTE (1) OF ADDRESS A4 IS SET TO A "1".
 BIT = 1 DEFINES TOTAL CONTROL OF REF BUG BY FLIGHT CONTROL SYSTEM ONLY, DISABLES LOCAL CONTROL; BIT = 0 ALLOWS LOCAL CONTROL OF REFERENCE BUG.

ADDRESS 73 TITLE: +/- DELTA TORQUE REQUEST & PWT STATUS

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[0	1	1	1]	[0	0	1	1]
	[-----HEX 7 -----]				[-----HEX 3 -----]			
	[-----ADDRESS BYTE-----]							
(1)	PWT VALID 1=VAL	PWT ACTIVE 1=ACT	PLA SET 1=SET	GA MODE 1=GA	PAD	PAD	SI SOURCE IDENT	SI IDENT
		(NOTE 2)	(NOTE 3)				(NOTE 1)	
(2)	<-----2'S COMPLEMENT, 16-BIT-----LSB>							
	<-----PERCENT MAX TURBINE TORQUE----->							
	<----- +/- DELTA TORQUE REQUEST----->							
(3)	[MSB-----2/S COMPLEMENT, 16-BIT-----<							
	[-64% -----PERCENT MAX TURBINE TORQUE-----<							
	[DOWN----- +/- DELTA TORQUE REQUEST-----<							
(4)	[PAD	PAD	PAD	PAD	PAD	PAD	PAD	PAD]
	[-----SPARE-----]							
(5)	[PAD	PAD	PAD	PAD	PAD	PAD	PAD	PAD]
	[-----SPARE-----]							
(6)	[PAD	PAD	PAD	PAD	PAD	PAD	PAD	PAD]
	[-----SPARE-----]							
(7)	[PAD	PAD	PAD	PAD	PAD	PAD	PAD	PAD]
	[-----SPARE-----]							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

NOTE 2: PWT = POWER TRIM

NOTE 3: PLA = POWER LEVER ANGLE
 PLA SET IS A "1" WHEN THE FADEC DETERMINES POWER LEVERS ARE
 CORRECTLY SET FOR A POWER TRIM (PWT) OPERATION.

TITLE: PRESSURE ALTITUDE/BARO ALTITUDE
ADDRESS A0

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	0	1	0]	[0	0	0	0]
	[-----HEX A-----]				[-----HEX 0-----]			
	[-----ADDRESS BYTE-----]							
(1)	ALT PRES 1=VAL	ALT BARO 1=VAL	RSVD PRE SEL (NOTE 3)	STRUT SW 1=GND	BARO ALT 1=XSIDE (NOTE 2)	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----FEET-----]							
	◀-----ALTITUDE (PRESSURE)-----]							
(3)	[MSB----- (BIASED +24576 FEET, NOTE 4) -----◀-							
	[-32768-----FEET-----◀-							
	[-----ALTITUDE (PRESSURE) -----◀-							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----FEET-----]							
	◀-----ALTITUDE (BARO) -----]							
(5)	[MSB----- (BIASED +24576 FEET, NOTE 4) -----◀-							
	[-32768-----FEET-----◀-							
	[-----ALTITUDE (BARO) -----◀-							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED OR NOT DEFINED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

NOTE 2: BARO ALT BIT (3) BYTE (1)

THIS BIT = 1 INDICATES THE BAROMETRIC BASED DATA ONLY (BYTES 4 & 5) IS FROM THE SIDE OPPOSITE TO THE SIDE INDICATED BY SI BITS (1,0) BYTE (1). THIS BIT SHOULD ALWAYS BE SET THE SAME IN LABELS A0, A7, & A8.

NOTE 3: BIT (5) OF BYTE (1) IS BIT 1 OF A TWO-BIT MATRIX FOR ALTITUDE PRESELECT FUNCTION. BIT 2 IS ON ADDRESS A1.

NOTE 4: DECIMAL BIT WEIGHTS IN THIS FIELD MUST BE BIASED BY +24576 FEET FOR ALTITUDE VALUE.
EXAMPLE: 0 FEET OF ALTITUDE = DECIMAL VALUE OF -24576 IN THIS DATA FIELD

TITLE: VERT SPEED/IAS

ADDRESS A1

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1 0 1 0]				[0 0 0 1]			
	[-----HEX A-----]				[-----HEX 1-----]			
	[-----ADDRESS BYTE-----]							
(1)	VERT SPD 1=VAL	IAS 1=VAL	RSVD PRE SEL	ADC VALID 1=VAL	PAD	TEST 1=ON	SI SOURCE IDENT	SI (NOTE 1)
			(NOTE 3) (NOTE 2)					
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB] ◀-----FEET/MINUTE-----] ◀-----VERTICAL SPEED-----]							
(3)	[MSB-----] [-20480-----FEET/MINUTE-----] [DOWN-----VERTICAL SPEED-----]							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB] ◀-----KNOTS-----] ◀-----INDICATED AIRSPEED-(IAS)-----]							
(5)	[MSB-----] [-512-----KNOTS-----] [-----INDICATED AIRSPEED-(IAS)-----]							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED OR NOT DEFINED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

NOTE 2: ADC VALID MAY CONTAIN PRESELECTOR "KNOB IN MOTION" INFORMATION IN SOME AIR DATA COMPUTERS. ALTITUDE PRESELECTOR "KNOB IN MOTION" WILL FORCE THIS BIT TO ZERO (0) FOR THESE AIR DATA COMPUTERS.

NOTE 3: BIT (5) OF BYTE (1) IS BIT 2 OF A TWO-BIT MATRIX FOR ALTITUDE PRESELECT FUNCTION. BIT 1 IS ON ADDRESS A0.

ADDRESS A2 TITLE: TAS/ISA DELTA TEMP

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	0	1	0]	[0	0	1	0]
	[-----HEX A-----]				[-----HEX 2-----]			
	[-----ADDRESS BYTE-----]							
(1)	TAS	ISA	PAD	STRUT	PAD	TEST	SI	SI
				SW			SOURCE	
	1=VAL	1=VAL		1=GND		1=ON	IDENT	(NOTE 1)
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----KNOTS-----]							
	◀-----TRUE AIRSPEED--(TAS)-----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀							
	[-1024-----KNOTS-----◀							
	[-----TRUE AIRSPEED--(TAS)-----◀							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES C-----]							
	◀-----ISA DELTA TEMP-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀							
	[-512-----DEGREES C-----◀							
	[-----ISA DELTA TEMP-----◀							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED OR NOT DEFINED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

TITLE: MACH

ADDRESS A3

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	0	1	0]	[0	0	1	1]
	[-----HEX A-----]				[-----HEX 3-----]			
	[-----ADDRESS BYTE-----]							
(1)	MACH DSPLY 1=VAL	MACH TRUE 1=VAL	PAD	MACH DSPLY 1=TRUE	PAD	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	←-----2'S COMPLEMENT, 16-BIT-----LSB] ←----- ←-----MACH NUMBER (M) -- (DISPLAYED) -----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----← [-1.024-----← [-----MACH NUMBER (M) - (DISPLAYED) -----←							
(4)	←-----2'S COMPLEMENT, 16-BIT-----LSB] ←----- ←-----MACH NUMBER (M) -- (TRUE) -----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----← [-1.024-----← [-----MACH NUMBER (M) -- (TRUE) -----←							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED OR NOT DEFINED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

TITLE: VS/IAS BUGS

ADDRESS A4

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	0	1	0]	[0	1	0	0]
	[-----	HEX A-----	[-----	HEX 4-----	[-----	ADDRESS BYTE-----	[-----	]
(NOTE 2)								
(1)	VS	IAS	VS	IAS	VS/IAS	TEST	SI	SI
	REF	BUGS	REF	BUGS	STATUS		SOURCE	
	[1=DSPLY ON]	[1=ADJUST ON]	1=ON	1=ON	IDENT			
(2)	←-----	2'S COMPLEMENT, 16-BIT-----	←-----	LSB]	←-----	FEET/MINUTE-----	←-----	]
	←-----	VERTICAL SPEED	BUG-----	←-----				
(3)	[MSB-----	2'S COMPLEMENT, 16-BIT-----	←-----	]	[-----	-20480-----	FEET/MINUTE-----	←-----
	[DOWN-----	VERTICAL SPEED	BUG-----	←-----				
(4)	←-----	2'S COMPLEMENT, 16-BIT-----	←-----	LSB]	←-----	KNOTS-----	←-----	]
	←-----	INDICATED AIRSPEED	BUG-----	←-----				
(5)	[MSB-----	2'S COMPLEMENT, 16-BIT-----	←-----	]	[-----	-512-----	KNOTS-----	←-----
	[-----	INDICATED AIRSPEED	BUG-----	←-----				

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

THESE BITS ARE RE-TRANSMITTED FROM THE SI BITS OF LABEL 72 IN ORDER TO IDENTIFY THE ORIGINAL SOURCE OF REFERENCE DATA AS FOLLOWS:

1	0	
0	0	NOT USED OR NOT DEFINED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

IF NO VALID LABEL 72 IS RECEIVED, THESE BITS WILL BE TRANSMITTED AS 0,0.

NOTE 2: VS/IAS STATUS BIT (3) BYTE (1)

BIT = 1 INDICATES BOTH VS AND IAS BUG DATA IS AVAILABLE FULL TIME;
BIT = 0 INDICATES BUG DATA ORIGINATES IN FLIGHT CONTROL SYSTEM AND IS NOT FULL TIME. BUG DATA DEPENDS ON FLIGHT CONTROL SELECTION OF MODES USING VS OR IAS REFERENCES.

ADDRESS A6

TITLE: SAT, V_{MO}/M_{MO}

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1 0 1 0]				[0 1 1 0]			
	[-----HEX A-----]				[-----HEX 6-----]			
	[-----ADDRESS BYTE-----]							
(1)	SAT	V_{MO} / M_{MO}	OVERSPEED		PAD	TEST	SI	SI
	1=VAL	1=VAL	1=ON	1=ON		1=ON	SOURCE IDENT	
							(NOTE 1)	
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES C-----]							
	◀-----STATIC AIR TEMP (SAT)-----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----LSB]							
	[-512-----DEGREES C-----LSB]							
	[-----STATIC AIR TEMP (SAT)-----LSB]							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----KNOTS-----]							
	◀-----MAX ALLOWABLE AIRSPEED (V_{MO}/M_{MO})-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----LSB]							
	[-512-----KNOTS-----LSB]							
	[-----MAX ALLOWABLE AIRSPEED (V_{MO}/M_{MO})-----LSB]							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED OR NOT DEFINED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

TITLE: TAT/PRESEL ALTITUDE

ADDRESS A7

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	0	1	0]	[0	1	1	1]
	[-----HEX A-----]				[-----HEX 7-----]			
	[-----ADDRESS BYTE-----]							
(1)	TAT	PRE	ALT	RSVD	PRE	TEST	SI	SI
	ALT	ALT	ALERT	ALTS	SEL		SOURCE	
	1=VAL	1=VAL	1=ALERT	KNOB	1=XSIDE	1=ON	IDENT	
					(NOTE 2)		(NOTE 1)	
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES C-----]							
	◀-----TOTAL AIR TEMP--- (TAT)-----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀							
	[-512-----DEGREES C-----◀							
	[-----TOTAL AIR TEMP--- (TAT)-----◀							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----FEET-----]							
	◀-----ALTITUDE (PRE-SELECTOR)-----]							
(5)	[MSB----- (BIASED +24576 FEET, NOTE 3) -----◀							
	[-32768-----FEET-----◀							
	[-----ALTITUDE (PRE-SELECTOR)-----◀							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED OR NOT DEFINED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

NOTE 2: BARO ALT BIT (3) BYTE (1)

THIS BIT = 1 INDICATES THE BAROMETRIC BASED DATA ONLY (BYTES 4 & 5) IS FROM THE SIDE OPPOSITE TO THE SIDE INDICATED BY SI BITS (1,0) BYTE (1). THIS BIT SHOULD ALWAYS BE SET THE SAME IN LABELS A0, A7, & A8.

NOTE 3: DECIMAL BIT WEIGHTS IN THIS FIELD MUST BE BIASED BY +24576 FEET FOR ALTITUDE VALUE.
 EXAMPLE: 0 FEET OF ALTITUDE = DECIMAL VALUE OF -24576
 IN THIS DATA FIELD

ADDRESS A8

TITLE: M_{MO}/BARO SET

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1 0 1 0]				[1 0 0 0]			
	[-----HEX A-----]				[-----HEX 8-----]			
	[-----ADDRESS BYTE-----]							
(1)	M _{MO}	BARO	IN HG/ MB SEL	PAD	BARO SET	TEST	SI	SI
	1=VAL	1=VAL	1=MB		1=XSIDE	1=ON	SOURCE IDENT	
					(NOTE 2)		(NOTE 1)	
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB] ◀-----MACH UNITS-----] ◀-----M _{MO} -----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀ [-1.024-----MACH UNITS-----◀ [-----M _{MO} -----◀							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB] ◀-----INCHES OF MERCURY/MILLIBARS-----] ◀-----BARO SETTING-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀ [-32.0/-1084-----INCHES OF MERCURY/MILLIBARS-----◀ [-----BARO SETTING-----◀							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

```

1 0
0 0 NOT USED OR NOT DEFINED
0 1 #1 UNIT
1 0 #2 UNIT
1 1 #3 UNIT

```

NOTE 2: BARO ALT BIT (3) BYTE (1)

THIS BIT = 1 INDICATES THE BAROMETRIC BASED DATA ONLY (BYTES 4 & 5) IS FROM THE SIDE OPPOSITE TO THE SIDE INDICATED BY SI BITS (1,0) BYTE (1). THIS BIT SHOULD ALWAYS BE SET THE SAME IN LABELS A0, A7, & A8.

TITLE: FCS ANNUNCIATIONS

ADDRESS AA

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	0	1	0]	[1	0	1	0]
	[-----HEX A-----]				[-----HEX A-----]			
	[-----ADDRESS BYTE-----]							
(1)	[GLIDESLOPE		] [NAV-1		] SPD		[ALT PRESELECT]	
	[EXT CAP		ARM]		[CAP ARM]		HOLD [CAP/TRK ARM]	
	[-----FCS ANNUNCIATIONS-----]							
(2)	APP	APP SYNC	REPORT	ALT	IAS	MACH	TST	
	CAP	ARM SWITCH	CODE BAD	HOLD	HOLD	HOLD		
	[-----FCS ANNUNCIATIONS-----]							
(3)	VS	CLI	GO ARND	PTCH	ROLL	VNAV		
			PTCH	ROLL		CAP	ARM	
	[-----FCS ANNUNCIATIONS-----]							
(4)	BC	DR	HEADING	TRB	XFR	DES-	HLF	
			SEL	HLD		CENT		
	[-----FCS ANNUNCIATIONS-----]							
(5)	NAV-2	UNUSED	ARM	ALT PRESELECT	XCHAN	CERT		
	CAP	ARM	WARN	CAP	TRK	SYNC		
	[-----FCS ANNUNCIATIONS-----]							
(6)	RADIO	OUT OF TRIM	DISCONNECT	TRIM FAIL				
	CPLE	RUD	AIL	ELE	YD	AP	PITCH	YAW
	[-----FCS ANNUNCIATIONS-----]							
(7)	SPARE	SPARE	SPARE	SPARE	SPARE	NAC	V-BAR	STR
						TST	INVUE	VLD
	[-----FCS ANNUNCIATIONS-----]							

NOTE 1: A LOGIC "1" IN ANY BIT INDICATES THAT MODE IS "ON".

TITLE: FCS MODES

ADDRESS AB

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1 0 1 0] [-----HEX A-----] [-----ADDRESS BYTE-----] [1 0 1 1] [-----HEX B-----]							
(1)	HDG FD STDBY ROLL PITCH AIRCRAFT BUG ANNUN MODE [---STEER---] WEIGHT 1=SYNC DSBL 1=ON [1=INVVIEW] [HVY MED LIT] (NOTE 2) 1=DSBL							
(2)	[-----BIT CODE IS TBD-----] [-----TBD-----] [-----FCS SPARES-----]							
(3)	[-----BIT CODE IS TBD-----] [-----TBD-----] [-----FCS SPARES-----]							
(4)	[-----BIT CODE IS TBD-----] [-----TBD-----] [-----FCS SPARES-----]							
(5)	[-----BIT CODE IS TBD-----] [-----TBD-----] [-----FCS SPARES-----]							
(6)	LEVEL] ALT [---BIT CODE IS TBD---] ARM [-----TBD-----] <-----] [-----FCS SPARES-----]							
(7)	REV- GA ENGAGE SOFT DES- TRIM A/P ERSE Y/D A/P RIDE CENT RUN FAIL [-----FCS MODES-----]							

NOTE 1: A LOGIC "1" IN ANY BIT INDICATES THAT MODE IS "ON".

NOTE 2: HDG BUG SYNC BIT 7 BYTE (1) GOES TO A LOGIC "1" FOR A NOMINAL 1-SECOND TO REQUEST SYNC ACTION.

TITLE: SECONDARY IAS/MACH REFERENCE MARKERS

ADDRESS AC

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	0	1	0]	[1	1	0	0]
	[-----HEX A-----]				[-----HEX C-----]			
	[-----ADDRESS-----]				[-----BYTE-----]			
(1)	MARKERS		IAS/	PAD	PAD	TEST	SI	SI
	DSPLY	ADJ	MACH				SOURCE	
	[1=ON	1=ON]	1=MACH			1=ON	IDENT	(NOTE 1)
(2)	MKR	[MARKER SELECTION]			PAD	[MARKER COLOR]		
	TYPE	[(NOTE 2)]				[(NOTE 3)]		
	(NOTE 4)	[4	2	1]		[4	2	1]
(3)	[PAD	PAD	PAD	PAD	PAD	PAD	PAD	PAD]
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----KNOTS/MACH #-----]							
	◀-----IAS/MACH REFERENCE MARKER-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀--							
	[-512/-1.024-----KNOTS/MACH #-----◀--							
	[-----IAS/MACH REFERENCE MARKER-----◀--							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED OR NOT DEFINED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

NOTE 2: SELECTION BITS
6,5,4 OF BYTE (2)

4	2	1	
0	0	0	VR
0	0	1	VF
0	1	0	V1
0	1	1	V2
1	0	0	VREF
1	0	1	MARKER A
1	1	0	MARKER B
1	1	1	FCS MACH REF.

NOTE 3: COLOR BITS
2,1,0 OF BYTE (2)

4	2	1	
0	0	0	BLACK
0	0	1	BLUE
0	1	0	GREEN
0	1	1	CYAN
1	0	0	RED
1	0	1	MAGENTA
1	1	0	YELLOW
1	1	1	WHITE

NOTE 4: MKR TYPE BIT
BIT 7 OF BYTE (2)

0=BUG
1=LINE OR OTHER ALTERNATE SYMBOL

TITLE: LOW SPEED CUE AND SECONDARY IAS/MACH RANGE MARKERS
ADDRESS AD

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	0	1	0]	[1	1	0	1]
	[-----HEX A-----]				[-----HEX D-----]			
	[-----ADDRESS-----]				[-----BYTE-----]			
(1)	MARKER DSPLY 1=ON	RSVD	IAS/ MACH 1=MACH	PAD	PAD	TEST 1=ON	SI SOURCE IDENT (NOTE 1)	SI
(2)	MKR TYPE (NOTE 4)	[RANGE MKR SELECTION] (NOTE 2)	PAD	[RANGE MKR COLOR] (NOTE 3)				
	[4	2	1]		[4	2	1]	
(3)	[MSB-----8-BIT BINARY-----LSB] [----- (NOTE 5) -----] [-----MULTIPLIER/RANGE MKR LENGTH-----]							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB] ◀-----KNOTS/MACH #-----] ◀-----V _s , MIN-MAX/IAS-MACH RANGE MKR TOP-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀- [-512/1.024-----KNOTS/MACH #-----◀- [-----V _s , MIN-MAX/IAS-MACH RANGE MKR TOP-----◀-							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

<u>1</u>	<u>0</u>	
0	0	NOT USED OR NOT DEFINED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

NOTE 2:
SELECTION BITS
(6,5,4) OF BYTE(2)

<u>4</u>	<u>2</u>	<u>1</u>	
0	0	0	Vs MIN
0	0	1	Vs MAX
0	1	0	Vs APPR MKR
0	1	1	VMKR1
1	0	0	VMKR2
1	0	1	VMKR3
1	1	0	VMKR4
1	1	1	VMKR5

NOTE 3:
COLOR BITS
(2,1,0) OF BYTE(2)

<u>4</u>	<u>2</u>	<u>1</u>	
0	0	0	BLACK
0	0	1	BLUE
0	1	0	GREEN
0	1	1	CYAN
1	0	0	RED
1	0	1	MAGENTA
1	1	0	YELLOW
1	1	1	WHITE

NOTE 4:
MARKER TYPE BIT
(7) OF BYTE(2)

- A. FOR MKR SELECTION
(BITS 6,5,4) OF
000 OR 001:
0=BAND OR RANGE
1=SINGLE VALUE
"TICK" MARK
- B. FOR MKR SELECTION
(BITS 6,5,4) OF 010:
0=DIAMOND APPR MKR
1="TICK" MARK
APPR MARKER

NOTE 5: DATA/BIT WEIGHTS FOR BYTE 3 ARE DETERMINED BY BYTE 2, BITS 6,5,4.
FOR 000, BYTE 3 IS NOMINAL 1.0 V_s MULTIPLIER WITH MSB=1.6
FOR 001 & 010, BYTE 3 IS NOMINAL 1.3 V_s MULTIPLIER WITH MSB=1.6.
FOR ALL OTHER CASES, BYTE 3 IS IAS/MACH RANGE MARKER LENGTH, WITH
MSB=128 KNOTS OR .128 MACH AND IS DEPENDENT ON BYTE 1, BIT 5.

TITLE: AIR DATA CONTROL & STATUS

ADDRESS AF

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
	-----	-----	-----	-----	-----	-----	-----	-----
(0)	[1 0 1 0]				[1 1 1 1]			
	[-----HEX A-----]				[-----HEX F-----]			
	[-----ADDRESS-----]				[-----BYTE-----]			
(1)	ALI FLAG 1=VAL	ASI FLAG 1=VAL	A/S REF CNTRL 1=ABS	OVRSPD WARN 1=TEST	RSVD SI2	TEST 1=ON	RSVD SI1	SI0
(2)	VS REF ON/OFF 1=CHNG	A/S REF ON/OFF 1=CHNG	ALT PRESEL 1=CNCL	BARO UNITS 1=MB	A/S REF INCR 1=CHNG	BARO STD 1=SET	RSVD A/S MODE	FL180 ANN 1=DSBL
(3)	[MSB-----2'S COMPLEMENT, 8-BIT-----LSB]							
	[-----INCREMENT/DECREMENT-----]							
	[-128-----BAROSET-----]							
(4)	[MSB-----2'S COMPLEMENT, 8-BIT-----LSB]							
	[-----INCREMENT/DECREMENT-----]							
	[-128-----ALTITUDE PRESELECT-----]							
(5)	[MSB-----2'S COMPLEMENT, 8-BIT-----LSB]							
	[-----INCREMENT/DECREMENT-----]							
	[-128-----A/S REF (IAS/MACH)-----]							
(6)	[MSB-----2'S COMPLEMENT, 8-BIT-----LSB]							
	[-----INCREMENT/DECREMENT-----]							
	[-128-----VERT SPEED REF-----]							
(7)	PAD	PAD	PAD	PAD	429 BUS	[ABS A/S REF SEL]		
					1=ABNORM	[BINARY POSITION]		
					(NOTE 1)	[4	2	1]

NOTE 1: BYTE (7) BIT 3 SET TO "1" INDICATES DISPLAY UNIT IS DISPLAYING
CROSS SIDE ALTIMETER INFORMATION.

TITLE: PITCH CMD/VERTICAL DEV

ADDRESS B0

```

BYTE      BIT      BIT      BIT      BIT      BIT      BIT      BIT
( # )      7        6        5        4        3        2        1        0
-----
(0)  [ 1          0          1          1 ] [ 0          0          0          0 ]
      [-----HEX B-----] [-----HEX 0-----]
      [-----ADDRESS BYTE-----]

(1)  PITCH      VERT      RSVD      VNAV      FCS      VNAV      (NOTE 1) (NOTE 2)
      CMD        DEV        VERT      STATE     VNAV      ENABL      LRU      DATA
      1=VAL      1=VAL      CLR      1=STBY    1=CAP      1=ON      IDENT

      (NOTE 3) (NOTE 4)

(2)  <-----2'S COMPLEMENT, 16-BIT-----LSB]
      <-----DEGREES-----]
      <-----PITCH COMMANDS (VNI/LRN)-----]

(3)  [MSB-----]
      [-90-----DEGREES-----]
      [FLY DOWN-----PITCH COMMANDS (VNI/LRN) -----]

(4)  <-----2'S COMPLEMENT, 16-BIT-----LSB]
      <-----FEET-----]
      <-----VNI/LRN DEVIATION-----]

(5)  [MSB-----2'S COMPLEMENT, 16-BIT-----]
      [-8192-----FEET-----]
      [FLY UP-----VNI/LRN DEVIATION-----]

```

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1) IS SET SAME AS BIT (0) OF ADDRESS C5 OR ADDRESS C8, WHICHEVER IS RECEIVED; IF NEITHER IS RECEIVED, THEN THIS BIT IS DEFAULTED TO=0

NOTE 3: FCS VNAV CAP, (BIT 3 BYTE 1)

THE PURPOSE OF THIS BIT IS TO SIGNAL THE VERTICAL CAPTURE POINT TO THE FLIGHT CONTROL SYSTEM. IT WILL BE PRESENT ONLY IF FCS VNAV SEL (ADDRESS B1 BIT 3 BYTE 1) IS A "1", AND IT WILL REMAIN A "1" FOR 5 REPETITIONS (500 MILLISECONDS).

NOTE 4: VNAV ENABLE, (BIT 2, BYTE 1)

THE PURPOSE OF THIS BIT IS TO PROVIDE INTERLOCK LOGIC TO THE FLIGHT CONTROL SYSTEM FOR VNAV MODE SELECTION. VNAV ENABLE = 1 INDICATES CONDITIONS ARE SUITABLE FOR ALLOWING SELECTION OF VNAV MODE ON THE FCS.

TITLE: DISTANCE TO TRK/ALT

ADDRESS B1

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	0	1	1]	[0	0	0	1]
	[-----HEX B-----]				[-----HEX 1-----]			
	[-----ADDRESS BYTE-----]							
(1)	DIST ALT 1=VAL	DIST TRK 1=VAL	OFF- SET 1=VAL	FCS VNAV 1=SEL	VNAV STATE 1=CAP	TEST 1=ON	(NOTE 1) LRU	(NOTE 2) DATA IDENT
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----NAUTICAL MILES-----]							
	◀-----DISTANCE TO ALTITUDE OR TRACK-----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀-							
	[-2048-----NAUTICAL MILES-----◀-							
	[-----DISTANCE TO ALTITUDE OR TRACK-----◀-							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----NAUTICAL MILES-----]							
	◀-----SELECTED OFFSET-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀-							
	[-2048-----NAUTICAL MILES-----◀-							
	[BEFORE-----SELECTED OFFSET-----]							

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1) IS SET SAME AS BIT (0) OF ADDRESS C8 IF RECEIVED; ELSE SAME AS BIT (1) ABOVE.

TITLE: SEL ANGLE/COMP ANGLE

ADDRESS B2

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1 0 1 1]				[0 0 1 0]			
	[-----HEX B-----]				[-----HEX 2-----]			
	[-----ADDRESS BYTE-----]							
(1)	SEL ANGLE 1=VAL	COMP ANGLE 1=VAL	PAD	PAD	ANGLE MODE 1=ARM	TEST 1=ON	(NOTE 1) LRU	(NOTE 2) DATA IDENT
(2)	◀-----2'S COMPLEMENT,16-BIT-----LSB] ◀-----DEGREES-----] ◀-----SELECTED VERTICAL ANGLE-----]							
(3)	[MSB-----2'S COMPLEMENT,16-BIT-----◀- [-90-----DEGREES-----◀- [-----SELECTED VERTICAL ANGLE-----◀-							
(4)	◀-----2'S COMPLEMENT,16-BIT-----LSB] ◀-----DEGREES-----] ◀-----COMPUTED VERTICAL ANGLE-----]							
(5)	[MSB-----2'S COMPLEMENT,16-BIT-----◀- [-90-----DEGREES-----◀- [-----COMPUTED VERTICAL ANGLE-----◀-							

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1) IS SET SAME AS BIT (0) OF ADDRESS C8 IF RECEIVED; ELSE SAME AS BIT (1) ABOVE.

TITLE: AIMPOINT ALT/VIS SELECT

ADDRESS B3

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	0	1	1]	[0	0	1	1]
	[-----	HEX B-----	[-----	HEX 3-----	[-----	HEX 3-----	[-----	HEX 3-----
	[-----	ADDRESS BYTE-----	[-----	ADDRESS BYTE-----	[-----	ADDRESS BYTE-----	[-----	ADDRESS BYTE-----
(1)	ALT AMPT 1=VAL	VS SEL 1=VAL	VNAV SCALE 1=ENABL	ELEV CORR (NOTE 3)	VS SEL MODE 1=ARM	TEST 1=ON	(NOTE 1) LRU	(NOTE 2) DATA IDENT
(2)	[-----	2'S COMPLEMENT, 16-BIT-----	[-----	LSB]-----	[-----	FEET-----	[-----	FEET-----
	[-----	ALTITUDE (AIMPOINT)-----	[-----	ALTITUDE (AIMPOINT)-----	[-----	ALTITUDE (AIMPOINT)-----	[-----	ALTITUDE (AIMPOINT)-----
(3)	[MSB-----	(BIASED +24576 FEET)-----	[-----	FEET-----	[-----	FEET-----	[-----	FEET-----
	[-----	ALTITUDE (AIMPOINT)-----	[-----	ALTITUDE (AIMPOINT)-----	[-----	ALTITUDE (AIMPOINT)-----	[-----	ALTITUDE (AIMPOINT)-----
(4)	[-----	2'S COMPLEMENT, 16-BIT-----	[-----	LSB]-----	[-----	FEET/MINUTE-----	[-----	FEET/MINUTE-----
	[-----	VERTICAL SPEED (SELECT)-----	[-----	VERTICAL SPEED (SELECT)-----	[-----	VERTICAL SPEED (SELECT)-----	[-----	VERTICAL SPEED (SELECT)-----
(5)	[MSB-----	FEET/MINUTE-----	[-----	FEET/MINUTE-----	[-----	FEET/MINUTE-----	[-----	FEET/MINUTE-----
	[DOWN-----	VERTICAL SPEED (SELECT)-----	[-----	VERTICAL SPEED (SELECT)-----	[-----	VERTICAL SPEED (SELECT)-----	[-----	VERTICAL SPEED (SELECT)-----

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1) IS SET SAME AS BIT (0) OF ADDRESS C8 IF RECEIVED; ELSE SAME AS BIT (1) ABOVE.

NOTE 3: 1=SLANT RANGE CORRECTION COMPUTED BY VNI.

TITLE: EFIS CONTROL & STATUS

ADDRESS C0

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1 1 0 0]				[0 0 0 0]			
	[-----HEX C-----]				[-----HEX 0-----]			
	[-----ADDRESS BYTE-----]							
(1)	EFIS STATUS 1=NORM	RSVD EFIS MODE	DRIVE LEFT 1=XFR	XFR RIGHT 1=XFR	PAD	TEST (NOTE 6) 1=ON	(NOTE 1) LRU	(NOTE 1) IDENT
(2)	[-----ACTIVE DEVIATION SOURCE SELECTED-----]				[-----DEV TYPE MATRIX-----]			
	[SEL SRCE ID]	[DEV SCALE]	[-----DEV TYPE MATRIX-----]					
	[NOTE 7]	[NOTE 2]	[NOTE 8]					
(3)	[-----PRESET DEVIATION SOURCE SELECTED-----]				[-----DEV TYPE MATRIX-----]			
	[SEL SRCE ID]	[DEV SCALE]	[-----DEV TYPE MATRIX-----]					
	[NOTE 7]	[NOTE 2]	[NOTE 8]					
(4)	L-NAV CLEAR PULSE	GS DSPLY 1=ON	VNAV DSPLY 1=ON	V-BAR /XPTR 1=V-B	FD DSPLY 1=ON	[MARKER BEACONS] [OUT MID INR]		
(5)	[V-NAV] [CLEAR] [PULSE]	[PAD [-----	PAD [-----	PAD [-----	PAD [-----	[-----CATEGORY II-----] [RSVD] [GRN-RED/AMB] [NOTE 9]		

NOTE 1: LRU IDENT BITS (1,0) BYTE (1) IDENTIFY BUS ORIGIN AS FOLLOWS:

```
DPU SETS BIT 1 BY SIDESTRAP, 1=SIDE 2 ; BIT 0 SET TO 0=DPU
MPU LEFT BUS SETS BIT 1 TO 0=SIDE 1 ; BIT 0 SET TO 1=MPU
MPU RIGHT BUS SETS BIT 1 TO 1=SIDE 2 ; BIT 0 SET TO 1=MPU
```

NOTE 2: NAV SCALE CODE, BYTES (2 & 3), BITS (5, 4)

$\frac{5}{0}$	$\frac{4}{0}$	UNDEFINED
0	1	1 MI/DOT
1	0	5 MI/DOT
1	1	10 MI/DOT

(NOTES CONTINUED ON FOLLOWING PAGE.)

TITLE: EFIS CONTROL & STATUS (continued)

ADDRESS C0 (continued)

NOTE 3: NAV PULSE, BIT (7) BYTE (4)

THIS BIT MUST CHANGE TO A (1) WHENEVER THE NAV DATA UNDER ADDRESSES C2 OR C4 CHANGE TO REPRESENT A NEW SOURCE. IT MUST REMAIN A "1" FOR NOT < 2 NOR > 5 CONSECUTIVE TRANSMISSIONS AND THEN RETURN TO A "0". INTERCHANGE OF (2) COURSES SHALL CAUSE A NAV PULSE, BUT TURNING A COURSE SELECT KNOB SHALL NOT. THIS PULSE IS NOT TO OCCUR IN THE EVENT OF A NAVIGATION SOURCE FAILURE FLAG OR DEAD DATA BUS INPUT TO THE EFIS.

NOTE 4: GS DISPLAY BIT (6) BYTE (4)

THIS BIT MUST REPRESENT THE DISPLAY STATUS OF THE GS FOR THE SAME SIDE AS THE DATA UNDER ADDRESS C4.

NOTE 5: DRIVE TRANSFER BITS (5,4) BYTE (1)

THESE BITS INDICATE WHEN DRIVE TRANSFER HAS BEEN SELECTED.

NOTE 6: TEST BIT (2) BYTE (1) INDICATES ONLY THAT EFIS IS IN TEST.

NOTE 7: SELECTED EHSI DEVIATION SOURCE IDENT BYTES (2,3) BITS (7,6)

$\frac{7}{0}$	$\frac{6}{0}$	NOT USED	$\frac{7}{1}$	$\frac{6}{0}$	#2 UNIT
0	1	#1 UNIT	1	1	#3 UNIT

NOTE 8: SELECTED EHSI DEVIATION SOURCE TYPE BYTES (2,3) BITS (3,2,1,0)

$\frac{3}{0}$	$\frac{2}{0}$	$\frac{1}{0}$	$\frac{0}{0}$	NOT USED	$\frac{3}{1}$	$\frac{2}{0}$	$\frac{1}{0}$	$\frac{0}{0}$	TACAN
0	0	0	1	VOR/LOC & GLS	1	0	0	1	SPARE
0	0	1	0	RESERVED	1	0	1	0	SPARE
0	0	1	1	LNAV & VNAV	1	0	1	1	SPARE
0	1	0	0	MAZ & MGP	1	1	0	0	SPARE
0	1	0	1	ADF	0	1	0	1	SPARE
0	1	1	0	GPS	0	1	1	0	SPARE
0	1	1	1	RNAV	1	1	1	1	SPARE

NOTE 9: CATEGORY II ANNUNCIATOR STATUS BYTE (5), BITS (1,0)

$\frac{1}{0}$	$\frac{0}{0}$	NOT SELECTED
0	1	FLASH YELLOW/RED
1	0	STEADY YELLOW/RED
1	1	STEADY GREEN

TITLE: HDG/HDG ERROR

ADDRESS C1

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	1	0	0]	[0	0	0	1]
	[-----HEX C-----]				[-----HEX 1-----]			
	[-----ADDRESS BYTE-----]							
(1)	HDG	HDG	MAG/	PAD	PAD	PAD	NOTE 1	NOTE 2
		ERR	TRUE				LRU	DATA
	1=VAL	1=VAL	1=TRUE				IDENT (NOTE 3)	
(2)	←-----2'S COMPLEMENT, 16-BIT-----LSB]							
	←-----DEGREES-----]							
	←-----HEADING (MAG/TRUE)-----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----←							
	[-180-----DEGREES-----←							
	[SOUTH-----HEADING (MAG/TRUE)-----←							
(4)	←-----2'S COMPLEMENT, 16-BIT-----LSB]							
	←-----DEGREES-----]							
	←-----HEADING ERROR-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----←							
	[-180-----DEGREES-----←							
	[STEER R-----HEADING ERROR=A/C HDG-SEL HDG-----←							

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1):

(A) STD EFIS: SET SAME AS MENU IF SELECTED: ELSE SAME AS BIT (1)

(B) SAAB EFIS:SET SAME AS BIT (1)

NOTE 3: THESE IDENT BITS DO NOT CHANGE IF CROSS-SIDE HEADING IS SELECTED FOR DISPLAY.

TITLE: CRS ERROR (ACTIVE/NEXT)

ADDRESS C2

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	1	0	0]	[0	0	1	0]
	[-----HEX C-----]				[-----HEX 2-----]			
	[-----ADDRESS BYTE-----]							
(1)	ACTV CRS 1=VAL	PRE CRS 1=VAL	PAD	PAD	PAD	PAD	(NOTE 1) LRU IDENT	(NOTE 2) DATA
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES-----]							
	◀-----ACTIVE COURSE ERROR=A/C HDG-SEL CRS-----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀-							
	[-180-----DEGREES-----◀-							
	[STEER R-----ACTIVE COURSE ERROR-----◀-							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES-----]							
	◀-----NEXT COURSE ERROR-(RESERVED)-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀-							
	[-180-----DEGREES-----◀-							
	[STEER R-----NEXT COURSE ERROR--(RESERVED)-----◀-							

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1):
 (A) STD EFIS: SET SAME AS MENU IF SELECTED: ELSE SAME AS BIT (1)
 (B) SAAB EFIS: SET SAME AS BIT (1)

NOTE 3: THESE IDENT BITS DO NOT CHANGE IF CROSS-SIDE HEADING IS SELECTED FOR DISPLAY.

NOTE 4: OLDER EQUIPMENT WILL CONTAIN DATA FOR "ACTV", BUT ONLY PAD BITS FOR "NEXT".

TITLE: RADIO ALT/DH

ADDRESS C3

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	1	0	0]	[0	0	1	1]
	[-----HEX C-----]				[-----HEX 3-----]			
	[-----ADDRESS BYTE-----]							
(1)	RAD ALT 1=VAL (NOTE 3)	ON SCALE 1=ON (NOTE 4)	DH 1=TRIP	PAD	PAD	RA TEST 1=ON (NOTE 5)	(NOTE 1) LRU	(NOTE 2) DATA IDENT
(2)	◀-----2'S COMPLEMENT, 16-BIT-----▶ ◀-----FEET-----▶ ◀-----RADIO ALTITUDE-----▶							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----▶ [-4096-----FEET-----▶ [-----RADIO ALTITUDE-----▶							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----▶ ◀-----FEET-----▶ ◀-----DECISION HEIGHT (DH) SET-----▶							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----▶ [-4096-----FEET-----▶ [-----DECISION HEIGHT (DH) SET-----▶							

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1):

- (A) STD EFIS: SET SAME AS MENU IF SELECTED: ELSE SAME AS BIT (1)
- (B) SAAB EFIS: SET SAME AS BIT (1)

NOTE 3: VALID, BIT (7) BYTE (1) REFLECTS ONLY THE STATE OF THE INPUT RAD ALT DISCRETE TO EFIS, AND IS GENERALLY A "1" ANYTIME THE RAD ALT IS POWERED UP.

NOTE 4: ON SCALE, BIT (6) BYTE (1) IS A NEW BIT GENERATED BY EFIS
AND INDICATES ONLY THE MAGNITUDE OF THE INCOMING ANALOG DATA.

NOTE 5: RA TEST, BIT (2) BYTE (1) INDICATES ONLY THAT RAD ALT IS IN TEST MODE, NOT ENTIRE EFIS BOX.

ADDRESS C4 TITLE: FCS LATERAL/VERTICAL DEVIATIONS (ACTIVE)

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	1	0	0]	[0	1	0	0]
	[-----	HEX C-----	[-----	HEX 4-----	[-----	ADDRESS BYTE-----	[-----	]
(1)	LAT DEV 1=VAL	VERT DEV 1=VAL	DEV TYPE 1=LIN	BACK LOC 1=ON	DELAYED ILS	PAD	(NOTE 1) LRU	(NOTE 2) DATA IDENT
			(NOTE 3)	(NOTE 5)				
(2)	◀-----	2'S COMPLEMENT, 16-BIT-(NOTE 4)	-----	LSB]	◀-----	DOTS (75 MV/DOT)	-----	]
	◀-----	LATERAL DEVIATION (ACTV)	-----	]	◀-----	MSB-----	-----	◀-----
(3)	[-8-----	DOTS (75 MV/DOT)	-----	◀-----	[FLY RT-----	LATERAL DEVIATION (ACTV)	-----	◀-----
(4)	◀-----	2'S COMPLEMENT, 16-BIT-(NOTE 4)	-----	LSB]	◀-----	DOTS (75 MV/DOT)	-----	]
	◀-----	VERTICAL DEVIATION (ACTV)	-----	]	◀-----	MSB-----	-----	◀-----
(5)	[-8-----	DOTS (75 MV/DOT)	-----	◀-----	[FLY UP-----	VERTICAL DEVIATION (ACTV)	-----	◀-----

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1):
 (A) STD EFIS: SET SAME AS MENU IF SELECTED: ELSE SAME AS BIT (1)
 (B) SAAB EFIS: SET BY NAV-L, NAV-R STRAP; ELSE SAME AS BIT (1)

NOTE 3: DEV TYPE=0 FOR LOC OR ANGULAR VOR

NOTE 4: RESOLUTION LIMITED TO 12 BITS FOR ANALOG INPUTS TO EFIS.

NOTE 5: DELAYED ILS, BIT (3) BYTE (1) REQUIRES LOC TO BE MENU SELECTED BY EFIS AS WELL AS NAV RCVR TUNED TO LOC TO BECOME A "1".

ADDRESS C5 TITLE: FCS LATERAL/VERTICAL DEVIATIONS (NEXT)

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	1	0	0]	[0	1	0	1]
	[-----HEX C-----]				[-----HEX 5-----]			
	[-----ADDRESS BYTE-----]							
(1)	LAT DEV 1=VAL	VERT DEV 1=VAL	DEV TYPE 1=LIN (NOTE 3)	BACK LOC 1=ON	DELAYED ILS (NOTE 5)	PAD	(NOTE 1) LRU	(NOTE 2) DATA IDENT
(2)	◀-----2'S COMPLEMENT, 16-BIT-(NOTE 4)-----LSB]							
	◀-----DOTS (75 MV/DOT)-----]							
	◀-----LATERAL DEVIATION (NEXT)-----]							
(3)	[MSB-----							
	[-8-----DOTS (75 MV/DOT)-----							
	[FLY RT-----LATERAL DEVIATION (NEXT)-----							
(4)	◀-----2'S COMPLEMENT, 16-BIT-(NOTE 4)-----LSB]							
	◀-----DOTS (75 MV/DOT)-----]							
	◀-----VERTICAL DEVIATION (NEXT)-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----							
	[-8-----DOTS (75 MV/DOT)-----							
	[FLY UP-----VERTICAL DEVIATION (NEXT)-----							

- NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2
- NOTE 2: DATA ORIGIN BIT (0) BYTE (1):
 (A) STD EFIS: SET SAME AS MENU IF SELECTED: ELSE SAME AS BIT (1)
 (B) SAAB EFIS: SET BY NAV-L, NAV-R STRAP; ELSE SAME AS BIT (1)
- NOTE 3: DEV TYPE=0 FOR LOC OR ANGULAR VOR
- NOTE 4: RESOLUTION LIMITED TO 12 BITS FOR ANALOG INPUTS TO EFIS.
- NOTE 5: DELAYED ILS, BIT (3) BYTE (1) REQUIRES LOC TO BE MENU SELECTED
 BY EFIS AS WELL AS NAV RCVR TUNED TO LOC TO BECOME A "1".
- NOTE 6: THIS ADDRESS IS USED BY CURRENT EQUIPMENT ONLY; OLDER
 EQUIPMENTS TRANSMIT IT AS ALL PAD BITS EXCEPT FOR BYTE (0).

TITLE: NAV ROLL/PITCH CMDS (ACTIVE)

ADDRESS C6

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0

(0)	[1	1	0	0]	[0	1	1	0]
	[-----HEX C-----]				[-----HEX 6-----]			
	[-----ADDRESS BYTE-----]							
(1)	ROLL CMD 1=VAL	PITCH CMD 1=VAL	COMMAND ROLL 1=YES	AVAIL PITCH 1=YES	REM CAP 1=CAP	REM ARM 1=ARM	(NOTE 1) LRU	(NOTE 2) DATA IDENT
	[NOTE 3]							
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES-----]							
	◀-----ROLL COMMANDS (ACTV)-----]							
(3)	[MSB-----◀							
	[-90-----DEGREES-----◀							
	[FLY LEFT-----ROLL COMMANDS (ACTV)-----◀							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DEGREES-----]							
	◀-----PITCH COMMANDS (ACTV)-----]							
(5)	[MSB-----◀							
	[-90-----DEGREES-----◀							
	[FLY DOWN-----PITCH COMMANDS (ACTV)-----◀							

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1):
 (A) STD EFIS: SET SAME AS MENU IF SELECTED: ELSE SAME AS BIT (1)
 (B) SAAB EFIS: SET BY NAV-L, NAV-R STRAP; ELSE SAME AS BIT (1)

NOTE 3: COMMAND AVAIL BITS(5,4) BYTE (1) WILL BE =1 ANYTIME COMMAND
 LABELS ARE RECEIVED ON THE LRN BUS. VALID_BITS (7,6)
 INDICATE WHETHER RECEIVED DATA IS VALID.

ADDRESS C7 TITLE: LOCALIZER/GLIDESLOPE DEVIATIONS

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1 1 0 0]				[0 1 1 1]			
	[-----HEX C-----]				[-----HEX 7-----]			
	[-----ADDRESS BYTE-----]							
(1)	LOC FLAG 1=VAL	GS FLAG 1=VAL	PAD	PAD	PAD	PAD	(NOTE 1) LRU	(NOTE 2) DATA IDENT
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DOTS (75 MV/DOT)-----]							
	◀-----LOCALIZER DEVIATION-----]							
(3)	[MSB-----NOTES 3 & 4 -----◀							
	[-8-----DOTS (75 MV/DOT)-----◀							
	[FLY RT-----LOCALIZER DEVIATION-----◀							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----DOTS (75 MV/DOT)-----]							
	◀-----GLIDESLOPE DEVIATION-----]							
(5)	[MSB-----NOTES 3 & 4 -----◀							
	[-8-----DOTS (75 MV/DOT)-----◀							
	[FLY UP-----GLIDESLOPE DEVIATION-----◀							

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1):
 (A) STD EFIS: SET SAME AS MENU IF SELECTED: ELSE SAME AS BIT (1)
 (B) SAAB EFIS: SET BY NAV-L, NAV-R STRAP; ELSE SAME AS BIT (1)
 (NAV-L, NAV-R IS IGNORED IF STRAPPED FOR FCC-86.)

NOTE 3: LOC/GS DEVIATIONS AT ALL TIMES WHETHER ACTIVE OR NOT,
 OR WHETHER DISPLAYED OR NOT.

NOTE 4: A NEGATIVE DEVIATION IS SENSED WHEN THE AIRCRAFT IS
 LEFT (BELOW) THE ILS, THUS REQUIRING A DEVIATION BAR DISPLAY
 TO THE RIGHT (ABOVE) THE AIRCRAFT SYMBOL, AND STEERING
 SIGNALS TO COMMAND RIGHT (UP) MOTION. REFER TO COMMENTS
 UNDER "SIGNS" PARAGRAPH ABOVE.

TITLE: FCS DISTANCE (ACTIVE/NEXT)

ADDRESS C8

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
	-----	-----	-----	-----	-----	-----	-----	-----
(0)	[1 1 0 0]				[1 0 0 0]			
	[-----HEX C-----]				[-----HEX 8-----]			
	[-----ADDRESS BYTE-----]							
(1)	ACTV DIST 1=VAL	NEXT DIST 1=VAL	ELEV CORR 1=ON	DME HOLD ACTV 1=ON	NEXT 1=ON	PAD	(NOTE 1) LRU	(NOTE 2) DATA IDENT
(2)	◀-----2'S COMPLEMENT,16-BIT-----▶ ◀-----NAUTICAL MILES-----▶ ◀-----ACTIVE-- DISTANCE-----▶							
(3)	[MSB-----2'S COMPLEMENT,16-BIT-----▶ [-2048---(NOTES 3 & 4)--NAUTICAL MILES-----▶ [-----ACTIVE-- DISTANCE-----▶							
(4)	◀-----2'S COMPLEMENT,16-BIT-----▶ ◀-----NAUTICAL MILES-----▶ ◀----- NEXT DISTANCE--(RESERVED)-----▶							
(5)	[MSB-----2'S COMPLEMENT,16-BIT-----▶ [-2048---(NOTES 3 & 4)--NAUTICAL MILES-----▶ [----- NEXT DISTANCE--(RESERVED)-----▶							

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1):
 (A) STD EFIS: SET SAME AS MENU IF SELECTED: ELSE SAME AS BIT (1)
 (B) SAAB EFIS:SET BY NAV-L, NAV-R STRAP; ELSE SAME AS BIT (1)

NOTE 3: THE CHANGE TO 2'S COMPLEMENT DID NOT CHANGE THE BIT WEIGHT.
 THE RANGE OF THE FORMAT IS THUS SHIFTED TO BE +/- 2048 NM,
 AND THE INCOMING DATA MAY EXCEED THAT RANGE. IF SO, THE
 DISTANCE DATA OUTPUT UNDER ADDRESS C8 MUST BE LIMITED SUCH
 THAT IT REMAINS SATURATED AND DOES NOT "ROLL OVER" IN THE
 2'S COMPLEMENT NOTATION AND CHANGE SIGN.

NOTE 4: OLDER EQUIPMENT WILL CONTAIN DATA FOR "ACTV", BUT ONLY
 PAD BITS FOR "NEXT".

TITLE: WIND VELOCITY/DIRECTION

ADDRESS C9

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	1	0	0]	[1	0	0	1]
	[-----HEX C-----]				[-----HEX 9-----]			
	[-----ADDRESS BYTE-----]							
(1)	WIND VALID 1=VAL	WIND AVAIL 1=YES (NOTE 3)	PAD	PAD	PAD	PAD	(NOTE 1) LRU	(NOTE 2) DATA IDENT
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB] ◀-----KNOTS-----] ◀-----WIND VELOCITY-----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀ [-2048-----KNOTS-----◀ [-----WIND VELOCITY-----◀							
(4)	◀-----2'S COMPLEMENT, 16-BIT-(NOTE 4)-----LSB] ◀-----DEGREES-----] ◀-----WIND DIRECTION-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-(NOTE 4)-----◀ [-180-----DEGREES-----◀ [-----WIND DIRECTION-----◀							

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1):
(A) STD EFIS: SET SAME AS MENU IF SELECTED; ELSE SAME AS BIT (1)
(B) SAAB EFIS: SET SAME AS BIT (1)

NOTE 3: WIND AVAIL, BIT (6) BYTE (1) WILL BE A "1" ANYTIME WIND DATA IS RECEIVED FROM THE LRN. VALIDITY IS INDICATED BY BIT (7).

NOTE 4: WIND DIRECTION IS THE ANGLE RELATIVE TO THE NOSE OF THE AIRCRAFT REGARDLESS OF THE FORMAT SENT FROM THE LRN., AND REGARDLESS OF THE COMPASS CARD DISPLAY AS MAG OR TRUE.

TITLE: SELECTED COURSE (ACTIVE/NEXT)

ADDRESS CA

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
	-----	-----	-----	-----	-----	-----	-----	-----
(0)	[1 1 0 0]				[1 0 1 0]			
	[-----HEX C-----]				[-----HEX A-----]			
	[-----ADDRESS BYTE-----]							
(1)	SEL COURSE		ACTIVE CRS		PAD	PAD	(NOTE 1)	(NOTE 2)
	ACTV	NEXT	TO	FROM			LRU	DATA
	1=VAL	1=VAL	1=TO	1=FROM			IDENT	
			(NOTE 4)				(NOTE 3)	
(2)	◀-----2'S COMPLEMENT, 16-BIT-(NOTE 5)-----LSB]							
	◀-----DEGREES-----]							
	◀-----SELECTED COURSE (ACTV)-----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-(NOTE 5)-----◀-							
	[-180-----DEGREES-----◀-							
	[SOUTH-----SELECTED COURSE (ACTV)-----◀-							
(4)	◀-----2'S COMPLEMENT, 16-BIT-(NOTE 5)-----LSB]							
	◀-----DEGREES-----]							
	◀-----SELECTED COURSE (NEXT)-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-(NOTE 5)-----◀-							
	[-180-----DEGREES-----◀-							
	[SOUTH-----SELECTED COURSE (NEXT)-----◀-							

NOTE 1: LRU IDENT BIT (1) BYTE (1): SET BY SIDESTRAP, 1=SIDE 2

NOTE 2: DATA ORIGIN BIT (0) BYTE (1):
 (A) STD EFIS: SET SAME AS MENU IF SELECTED: ELSE SAME AS BIT (1)
 (B) SAAB EFIS: SET BY NAV-L, NAV-R STRAP; ELSE SAME AS BIT (1)

NOTE 3: THESE IDENT BITS DO NOT CHANGE IF CROSS-SIDE HEADING IS
 SELECTED FOR DISPLAY.

NOTE 4: TO/FROM, BITS (5,4) BYTE (1) ARE CHANGED FROM 1 BIT TO 2
 BITS TO ALLOW RECOGNITION OF THE "NEITHER" STATE.

NOTE 5: OLDER EQUIPMENT WILL CONTAIN DATA FOR "ACTV", BUT ONLY
 PAD BITS FOR "NEXT".

EFIS JOYSTICK CURSOR POSITION								
ADDRESS CB (NOTE 1)								
BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
	-----	-----	-----	-----	-----	-----	-----	-----
(0)	[1 1 0 0] [1 0 1 1] [-----HEX C-----] [-----HEX B-----] [-----ADDRESS BYTE-----]							
(1)	CURSR CURSR 1=VAL	PAD PAD	PAD PAD	PAD PAD	PAD PAD	TEST TEST 1=ON	RESERVED SI SI IDENT	
(2)	<-----2'S COMPLEMENT, 16-BIT-----LSB] <-----NAUTICAL MILES-----] <-----DISTANCE TO JOYSTICK CURSOR-----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----<- [-2048-----NAUTICAL MILES-----<- [-----DISTANCE TO JOYSTICK CURSOR-----<-							
(4)	<-----2'S COMPLEMENT, 16-BIT-(NOTE 2)-----LSB] <-----DEGREES-----] <-----BEARING TO JOYSTICK CURSOR-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-(NOTE 2)-----<- [-180-----DEGREES-----<- [-----BEARING TO JOYSTICK CURSOR-----<-							

NOTE 1: THIS LABEL WILL BE TRANSMITTED AT A 10/SECOND RATE EACH TIME AN "ENTER" BUTTON IS PRESSED. OLDER EQUIPMENT WILL SEND A TOTAL OF (2) REPETITIONS; NEWER EQUIPMENT WILL SEND A TOTAL OF (20) REPETITIONS.

NOTE 2: BEARING IS RELATIVE TO NOSE OF AIRCRAFT

EFIS KEYSWITCH STATUS

ADDRESS CC (NOTE 1)

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	1	0	0]	[1	1	0	0]
	[-----	HEX C-----	[-----	HEX C-----	[-----	HEX C-----	[-----	HEX C-----
	[-----	ADDRESS BYTE-----	[-----	ADDRESS BYTE-----	[-----	ADDRESS BYTE-----	[-----	ADDRESS BYTE-----
(1)	KEY DATA 1=VAL	PAD	PAD	PAD	PAD	TEST 1=ON	SI SOURCE IDENT	SI
(2)	PAD	PAD	PAD	PAD	LEFT KEY	DOWN KEY	RIGHT KEY	UP KEY
	[-----	JOYSTICK KEYS, 1=	PRESSED	[-----	JOYSTICK KEYS, 1=	PRESSED	[-----	JOYSTICK KEYS, 1=
(3)	PAD	PAD	PAD	PAD	TOP KEY	2ND KEY	3RD KEY	BTM KEY
	[-----	RIGHT SIDE SELECT KEYS, 1=	PRESSED	[-----	RIGHT SIDE SELECT KEYS, 1=	PRESSED	[-----	RIGHT SIDE SELECT KEYS, 1=
(4)	[-----	RESERVED-----	[-----	RESERVED-----	[-----	RESERVED-----	[-----	RESERVED-----
	[-----	LEFT-SIDE SELECT KEYS, 1=	PRESSED	[-----	LEFT-SIDE SELECT KEYS, 1=	PRESSED	[-----	LEFT-SIDE SELECT KEYS, 1=
(5)	[-----	RESERVED-----	[-----	RESERVED-----	[-----	RESERVED-----	[-----	RESERVED-----
	[-----	BOTTOM SELECT KEYS, 1=	PRESSED	[-----	BOTTOM SELECT KEYS, 1=	PRESSED	[-----	BOTTOM SELECT KEYS, 1=

NOTE 1: THIS LABEL WILL BE TRANSMITTED AT A 10/SECOND RATE SO LONG AS ANY KEY IS PRESSED.

TITLE: HDG/HDG ERROR

ADDRESS CD

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	1	0	0]	[1	1	0	1]
	[-----HEX C-----]				[-----HEX D-----]			
	[-----ADDRESS BYTE-----]							
(1)	HDG BUG 1=VAL	HDG SOURCE 1=VAL	GA MODE 1=ON	FMS X-SIDE (NOTE 1)	PORT P/S	TEST 1=ON	EFIS SI (NOTE 2)	HDG SI
(2)	◀-----2'S COMPLEMENT, 16-BIT-(NOTE 3)-----LSB] ◀-----DEGREES-----] ◀-----SELECTED-HEADING (BUG) POSITION-----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-(NOTE 3)-----◀ [-180-----DEGREES-----◀ [SOUTH-----SELECTED-HEADING (BUG) POSITION-----◀							
(4)	◀-----2'S COMPLEMENT, 16-BIT-(NOTE 3)-----LSB] ◀-----DEGREES-----] ◀-----COMPASS HEADING-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-(NOTE 3)-----◀ [-180-----DEGREES-----◀ SOUTH-----COMPASS HEADING-----◀							

NOTE 1: ACTIVE FMS PORT BITS (4,3) BYTE (1):
 BIT (4): 1 = CROSS-SIDE PORT SELECTED
 BIT (3): 1 = PRIMARY PORT 0 = SECONDARY PORT

NOTE 2: EFIS SIDE IDENT BIT (1) BYTE (1): 1 = SIDE 2 0 = SIDE 1
HEADING SOURCE IDENT BIT (0) BYTE (1): 1= CROSS-SIDE

NOTE 3: COMPASS HEADING AND SELECTED HEADING ARE ALWAYS AS DISPLAYED

TITLE: MINIMUM & MAXIMUM STALL SPEEDS

ADDRESS CE

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
	-----	-----	-----	-----	-----	-----	-----	-----
(0)	[1 1 0 0]				[1 1 1 0]			
	[-----HEX C-----]				[-----HEX E-----]			
	[-----ADDRESS BYTE-----]							
(1)	ALPHA n VALID 1=VAL	IAS VALID 1=VAL	XSIDE ADC 1=XSIDE	XSIDE ALPHA n 1=XSIDE	PAD	PAD	[SI [SOURCE IDENT (NOTE 1)	SI]
(2)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----KNOTS IAS-----]							
	◀-----Vs MINIMUM (TOP OF RED)-----]							
(3)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀-							
	[-512-----KNOTS IAS-----◀-							
	[-----Vs MINIMUM (TOP OF RED)-----◀-							
(4)	◀-----2'S COMPLEMENT, 16-BIT-----LSB]							
	◀-----KNOTS IAS-----]							
	◀-----Vs MAXIMUM (TOP OF YELLOW)-----]							
(5)	[MSB-----2'S COMPLEMENT, 16-BIT-----◀-							
	[-512-----KNOTS IAS-----◀-							
	[-----Vs MAXIMUM (TOP OF YELLOW)-----◀-							

NOTE 1: SOURCE IDENT BITS (1,0) BYTE (1)

1	0	
0	0	NOT USED
0	1	#1 UNIT
1	0	#2 UNIT
1	1	#3 UNIT

TITLE: UNIVERSAL DIAGNOSTICS

GENERALIZED FORMAT

ADDRESS F3

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
(0)	[1	1	1	1]	[0	0	1	1]
	[-----HEX F-----]				[-----HEX 3-----]			
	[-----ADDRESS BYTE-----]							
(1)	DIAGNOSTICS CONDITION (NOTE 2)		PAD	FAULT DETCT 1=FLT	SI (NOTE 1)	SELF TEST 1=ON	SI SOURCE (NOTE 1)	SI IDENT
	(SEE NOTE 3)							
(2)	[0	0	0	0	0	0	0	0]
	[----- (8) 1-BIT MESSAGES, 0=NORMAL-----]							
	[-----DIAGNOSTIC MESSAGES, DEFINABLE FOR EACH BOX-----]							
(3)	[0	0	0	0	0	0	0	0]
	[----- (8) 1-BIT MESSAGES, 0=NORMAL-----]							
	[-----DIAGNOSTIC MESSAGES, DEFINABLE FOR EACH BOX-----]							
(4)	[0	0	0	0	0	0	0	0]
	[----- (8) 1-BIT MESSAGES, 0=NORMAL-----]							
	[-----DIAGNOSTIC MESSAGES, DEFINABLE FOR EACH BOX-----]							
	(SEE NOTE 4)							
	(SEE NOTE 5)							
(5)	[8	4	2	1]	[8	4	2	1]
	[-----BCD, 0-9 ONLY-----]				[-----BCD, 0-9 ONLY-----]			
	[-----2-DIGIT FAULT CODE, 00 TO 99-----]							

NOTE 1 : SOURCE IDENT BITS (3,1,0) BYTE (1)

3 1 0		3 1 0
0 0 0	NOT USED	1 0 0
0 0 1	#1 UNIT	1 0 1
0 1 0	#2 UNIT	1 1 0
0 1 1	#3 UNIT	1 1 1
		USED ONLY
		IF DEFINED
		FOR YOUR
		SPECIFIC BOX

(NOTES CONTINUE ON NEXT PAGE)

ADDRESS F3 (Continued)

NOTE 2: DIAGNOSTICS CONDITION IDENT BITS (7,6) BYTE (1)

THIS FIELD TO BE DEFINED BY THE BOX DESIGNER TO IDENTIFY ANY CONDITIONS ASSOCIATED WITH THE DIAGNOSTIC MESSAGE SUCH AS POWER-UP, MODE, ETC. IF MORE THAN 4 CONDITIONS ARE NEEDED, THE FIELD MAY BE EXPANDED TO INCLUDE BIT 5 AS WELL.

NOTE 3: THE 24 BITS IN BYTES 2,3,4 ARE TO BE DEFINED BY EACH BOX DESIGNER WITH THE ONLY RESTRICTION THAT ONCE DEFINED, A BIT MAY NOT CHANGE MEANING. UNUSED BITS ARE TRANSMITTED AS ZEROS, BUT MAY LATER BE USED IF THE DEFINITION IS EXPANDED. THE DIAGNOSTICS CONDITION BITS, BYTE 1, ALLOW FOR MULTIPLE USE OF THE 24 INDIVIDUAL BITS. "NORMAL" INFORMATION MAY BE DEFINED INTO THE 24-BIT FIELD IF IT WILL ASSIST THE DIAGNOSTICS PROCESS.

NOTE 4: BUSES WITH 8-BYTE FORMATS WILL HAVE (2) ADDITIONAL BYTES IN THE DIAGNOSTIC MESSAGE FIELD, MAKING A TOTAL OF (40) BITS AVAILABLE. THE FAULT CODE FIELD WILL REMAIN AT THE END, AND BECOMES BYTE (7).

NOTE 5: THE FAULT CODE FIELD IS INTENDED TO ALLOW A FAULT CONDITION TO BE PRIORITIZED, IF DESIRABLE, AND A SUITABLE MESSAGE DISPLAYED EITHER IN ADDITION TO OR INSTEAD OF THE NORMAL DIAGNOSTICS MEANS. EACH MESSAGE NUMBER WILL HAVE A MEANING TO BE DEFINED BY THE BOX DESIGNER.

ADDRESS F5 TITLE: UNIVERSAL ASCII PAGE DATA

BYTE (#)	BIT 7	BIT 6	BIT 5	BIT 4	BIT 3	BIT 2	BIT 1	BIT 0
	-----	-----	-----	-----	-----	-----	-----	-----
(0)	[1 1 1 1]				[0 1 0 1]			
	[-----HEX F-----]				[-----HEX 5-----]			
	[-----ADDRESS BYTE-----]							
(1)	[---BINARY,3-BIT---]				[-----BINARY,5-BIT-----]			
	[-----LINE-----]				[-----COLUMN-----]			
	[---IDENTIFIER---]				[-----IDENTIFIER-----]			
(2)	[-----8-BIT ASCII-----]							
	[----- (LEFT MOST) -----]							
	[-----CHARACTER #1-----]							
(3)	[-----8-BIT ASCII-----]							
	[-----CHARACTER #2-----]							
(4)	[-----8-BIT ASCII-----]							
	[-----CHARACTER #3-----]							
(5)	[-----8-BIT ASCII-----]							
	[-----CHARACTER #4-----]							

(NOTE 1)

NOTE 1: BUSES WITH 8-BYTE FORMATS WILL HAVE (2) ADDITIONAL BYTES,
 BYTES (6) AND (7), MAKING A TOTAL OF (6) CHARACTERS AVAILABLE.

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